

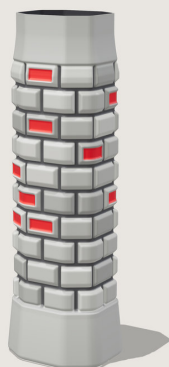
Pickleball Paddle Grips

Ten designs. One fitment.

Full specifications, true-size drawings and dimension data for every grip.



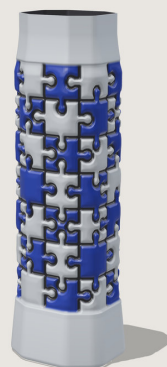
VORTEX GRIP
01 · 46.5 g PLA



CELLULAR MOD
02 · 47.7 g PLA



TESSEL-BLOCK
03 · 46.4 g PLA



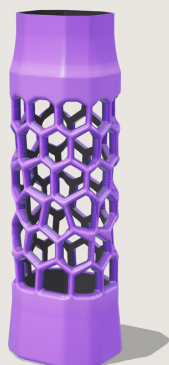
LOGIC-GRIP
04 · 48.0 g PLA



NEURO-TREAD
05 · 46.5 g PLA



CARBON MATRIX
06 · 48.1 g PLA



VORONOI-CORE
07 · 40.2 g PLA



TOPO-FLOW
08 · 47.8 g PLA



ERGO-CONTOUR
09 · 47.0 g PLA



HEXA-MOD
10 · 47.0 g PLA

132.82 mm

overall height

32.06 × 26.04 mm

bore, identical on all ten

40 – 49 g

solid, 100 % infill, any material

0.20 mm

layer height, no supports

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Click any entry to jump to the page.

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Weight, size and print-readiness compared

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What is identical on every grip, and the two deliberate deviations

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09 Ergo-Contour 37–40

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BACK MATTER

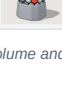
Notes, method and file manifest 45

HOW TO READ THESE SHEETS

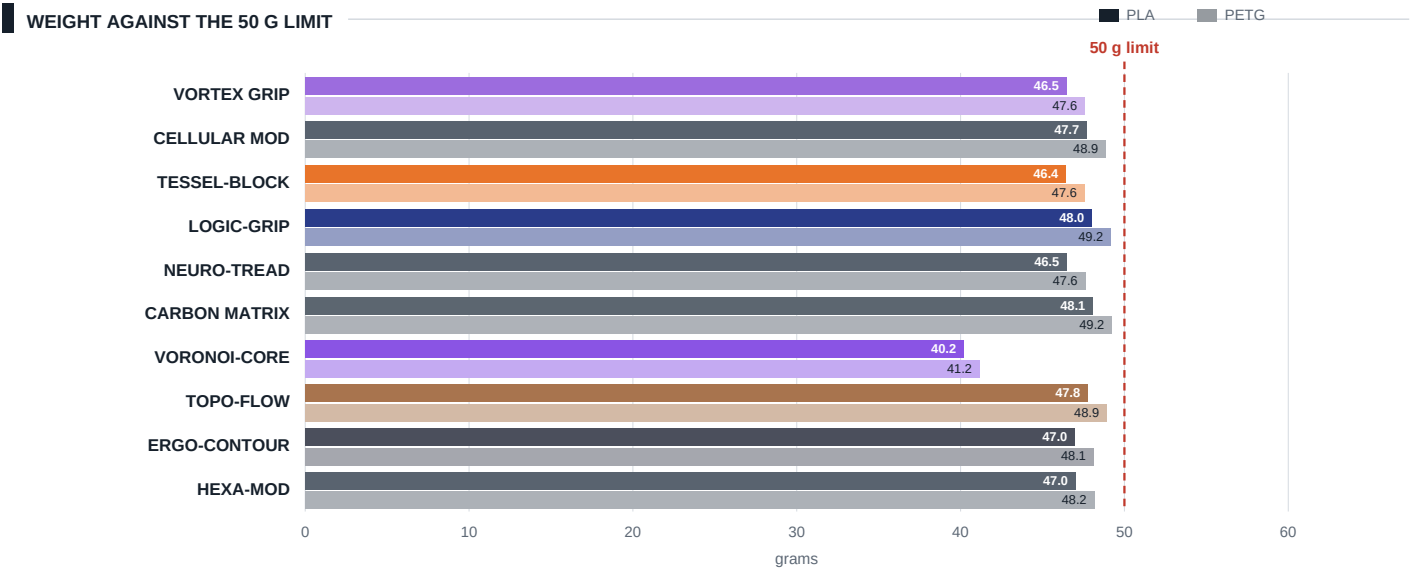
Every dimension is measured from the STL, not copied from the design intent. Drawings on the second page of each sheet are true size when printed at 100 %. Renders and colours are a preview; the STLs are single-material. Weight = volume × density at 100 % infill.

All ten at a glance

Same bore, same foot, same envelope; ten different surfaces, all under 50 g.

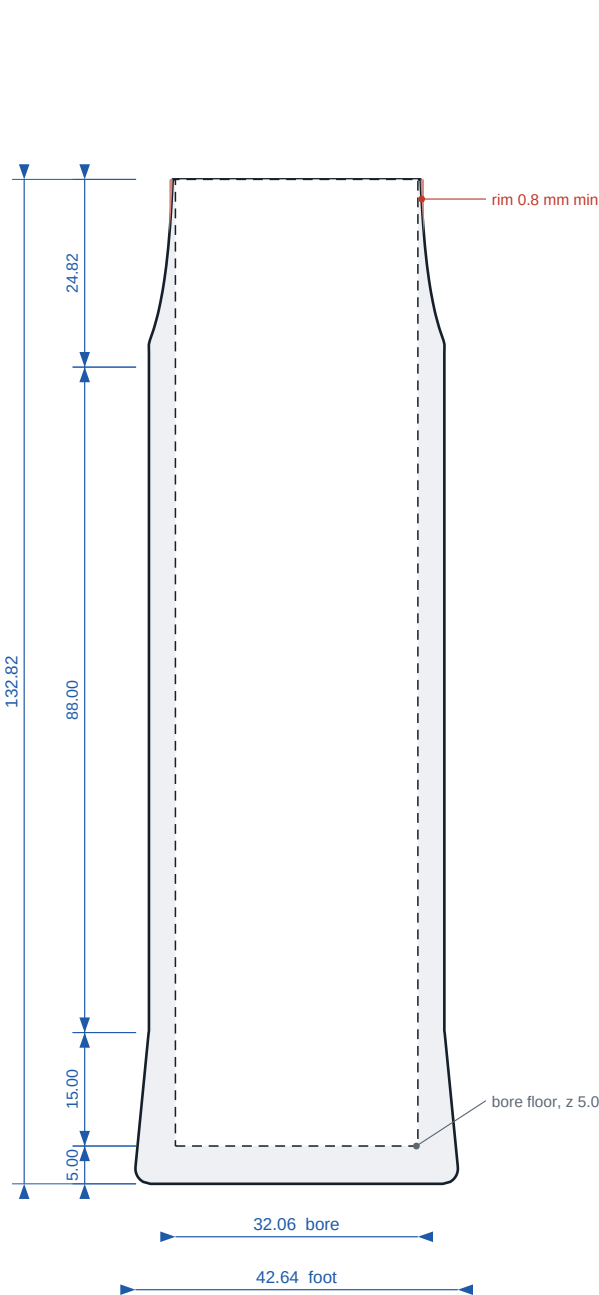
DESIGN	VOLUME	PLA g	PETG g	TPU g	ABS g	PEAK	BAND W×D	BRIDGE	COLOURS
 VORTEX GRIP PG-01	37.5 cm³	46.5	47.6	45.3	39.0	2.85	37.8 × 31.7	none	
 CELLULAR MOD PG-02	38.5 cm³	47.7	48.9	46.6	40.0	3.04	38.1 × 31.9	none	
 TESSEL-BLOCK PG-03	37.4 cm³	46.4	47.6	45.3	38.9	3.44	38.9 × 32.9	none	
 LOGIC-GRIP PG-04	38.7 cm³	48.0	49.2	46.9	40.3	2.81	37.7 × 31.7	none	
 NEURO-TREAD PG-05	37.5 cm³	46.5	47.6	45.4	39.0	3.50	39.1 × 33.0	none	
 CARBON MATRIX PG-06	38.8 cm³	48.1	49.2	46.9	40.3	2.73	37.5 × 31.4	none	
 VORONOI-CORE PG-07	32.4 cm³	40.2	41.2	39.2	33.7	3.50	39.1 × 33.0	3.8	
 TOPO-FLOW PG-08	38.5 cm³	47.8	48.9	46.6	40.1	3.13	38.3 × 32.1	none	
 ERGO-CONTOUR PG-09	37.9 cm³	47.0	48.1	45.8	39.4	2.77	36.9 × 30.9	none	
 HEXA-MOD PG-10	37.9 cm³	47.0	48.2	45.9	39.4	3.50	38.7 × 33.0	none	

Volume and weights: 100 % infill, volume × density (g/cm³: PLA 1.24, PETG 1.27, TPU 1.21, ABS 1.04). Peak = tallest texture above the bore, mm. Bridge = longest unsupported run, mm.

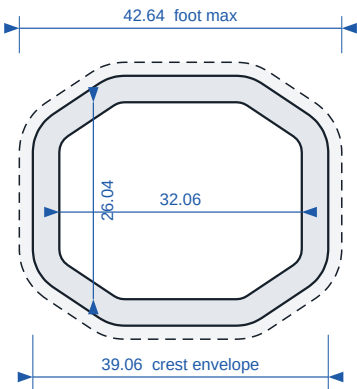


The reference fitment

Measured from PickleballGrip_1.3mf. Everything below is identical on all ten grips except where marked.



FRONT ELEVATION OF THE ENVELOPE (1 : 1)



PLAN: BORE, CREST ENVELOPE, FOOT (1 : 1)

IDENTICAL ON ALL TEN

Bore	32.06 × 26.04 mm, octagonal
Bore polygon	76 vertices, verbatim
Bore depth / floor	127.82 mm / z 5.0
Overall height	132.823 mm
Foot flare	z 0 – 20, max 42.63 × 36.61
Bottom fillet	about 2 mm, flat closed butt
Crest envelope	3.5 mm from the bore
Outer skin	never exceeds the reference

DELIBERATE DEVIATIONS

Rim wall ≥ 0.8 mm (reference tapers to 0.36 mm)
Under one nozzle line, would print as a lone thin line or be dropped.

Base wall 1.2 mm (reference 1.5 mm)
Three 0.4 mm perimeters; saves about 1.4 cm³ (1.7 g).

Texture heights 2.7 – 3.1 mm on six designs
Lowered plateaus keep every grip at or under 50 g.

ENVELOPE VS HEIGHT: REFERENCE AND GRIPS



HOW IT WAS VERIFIED

- Every STL is cross-sectioned at 40+ heights and compared with this profile.
- Bore: reference wall points must lie on the grip surface; nothing may enter the bore (0.000 mm²).
- Envelope: outer skin measured against the reference, worst case +0.001 mm.
- Print check: every 0.20 mm layer sliced; one closed body, no void, no material under 0.4 mm wide.

01

VORTEX GRIP

Braided over/under helical straps for dynamic multi-point contact.

46.5 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	37.8 × 31.7 mm
Peak texture height	2.85 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.47 cm³ / 318 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		46.5 g
PETG	1.27 g/cm³		47.6 g
TPU	1.21 g/cm³		45.3 g
ABS	1.04 g/cm³		39.0 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.6 m (100 % flow)

DESIGN INTENT

Sixteen straps wound in opposite directions cross in an over/under braid, leaving dark valleys between them. Every crossing gives the hand an edge to bite on, so the grip works from any hand position on the handle.

Straps rise no faster than 45 degrees, so no supports are needed.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 8 right-hand + 8 left-hand straps on a 36 degree helix
- Over/under weave: the strap on top alternates at every crossing
- Valleys drop to about the 1.2 mm base wall, which keeps the mass low

FILES

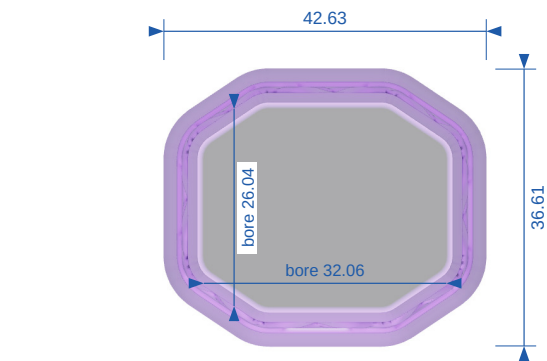
stl/01_vortex_grip.stl	19.9 MB · 398,860 triangles
3mf/01_vortex_grip.3mf	4.5 MB · colour

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

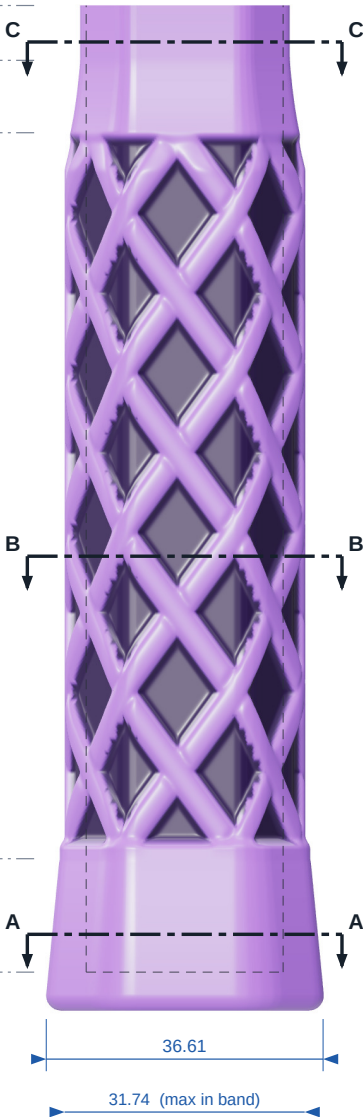
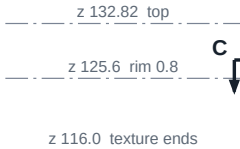
✓ 3MF VALIDATED



TOP VIEW (from +Z)



FRONT VIEW (from -Y)



RIGHT VIEW (from +X)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	37.76 × 31.74
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	2.85
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.

SCALE 1 : 1



PART

VORTEX GRIP

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

37.47 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

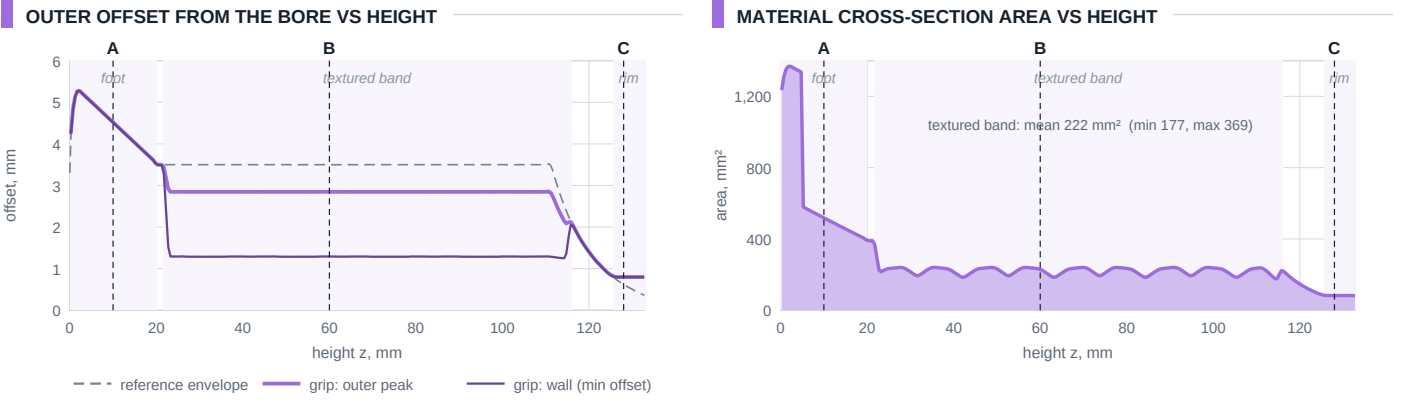
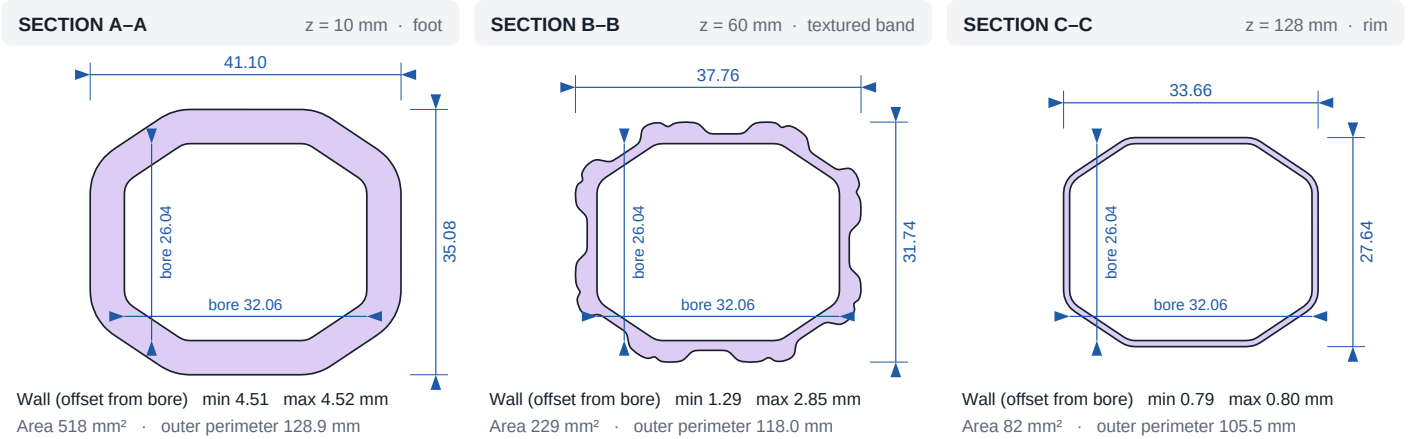
318 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	37.76 × 31.74 mm
Circumference in band	113.2 – 118.0 mm
Volume / surface area	37.47 cm ³ / 317.5 cm ²

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.29 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			2.85 mm

OUTER DIMENSIONS BY HEIGHT

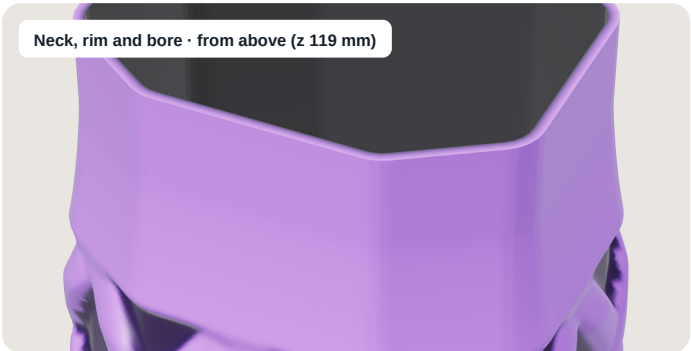
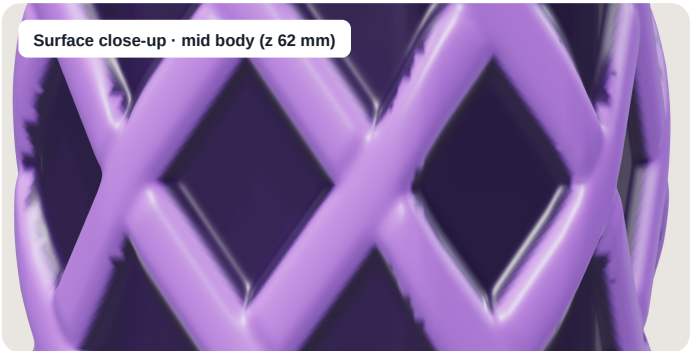
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	37.76	31.74	114.5	215	1.29	2.85
45.0	37.76	31.74	118.0	225	1.29	2.85
60.0	37.76	31.74	118.0	229	1.29	2.85
75.0	37.76	31.74	114.1	209	1.29	2.85
90.0	37.76	31.74	117.1	238	1.29	2.85
105.0	37.75	31.74	117.1	187	1.29	2.85
116.0	36.24	30.22	113.4	221	2.04	2.09
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

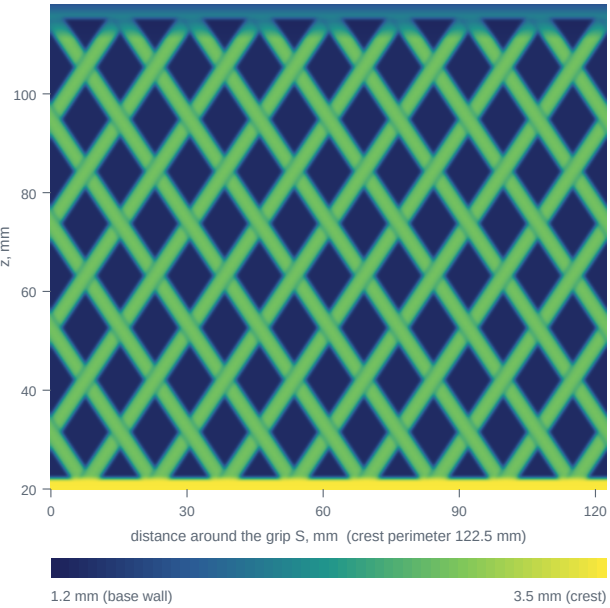
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Straps	8 + 8, counter-wound
Helix angle	36 deg from vertical
Strap pitch (around)	15.31 mm
Strap width	6.1 mm around / 5.0 mm across
Strap top height	2.35 – 2.86 mm above bore
Valley floor	1.29 mm above bore
Over/under step	0.37 mm

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

02

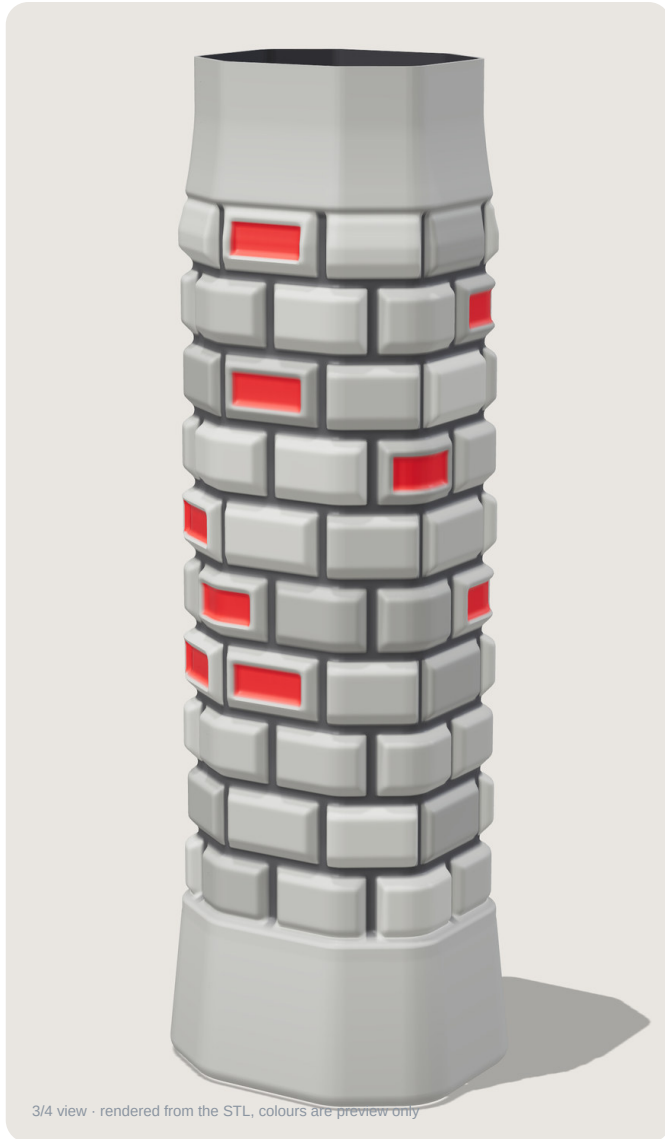
CELLULAR MOD

Staggered stacked cells; red cells are recessed pockets for swappable inserts.

47.7 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	38.1 × 31.9 mm
Peak texture height	3.04 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	38.47 cm ³ / 323 cm ²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm ³		47.7 g
PETG	1.27 g/cm ³		48.9 g
TPU	1.21 g/cm ³		46.6 g
ABS	1.04 g/cm ³		40.0 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 16.0 m (100 % flow)

DESIGN INTENT

A wall of staggered, individually pillowed cells. Roughly one cell in five is recessed as an insert pocket (red in the colourway), sized to take a contrasting insert. No insert parts are included.

Cell edges are chamfered; nothing overhangs beyond 45 degrees.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 10 rows × 8 staggered cells with 1.4 mm gaps
- Three cell heights give a subtly stepped surface
- About 22% of cells are recessed 0.7 – 1.2 mm as insert pockets

FILES

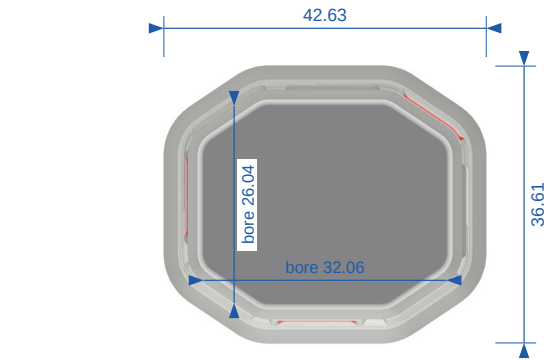
stl/02_cellular_mod.stl	14.7 MB · 294,052 triangles
3mf/02_cellular_mod.3mf	3.3 MB · colour

✓ FITMENT VERIFIED

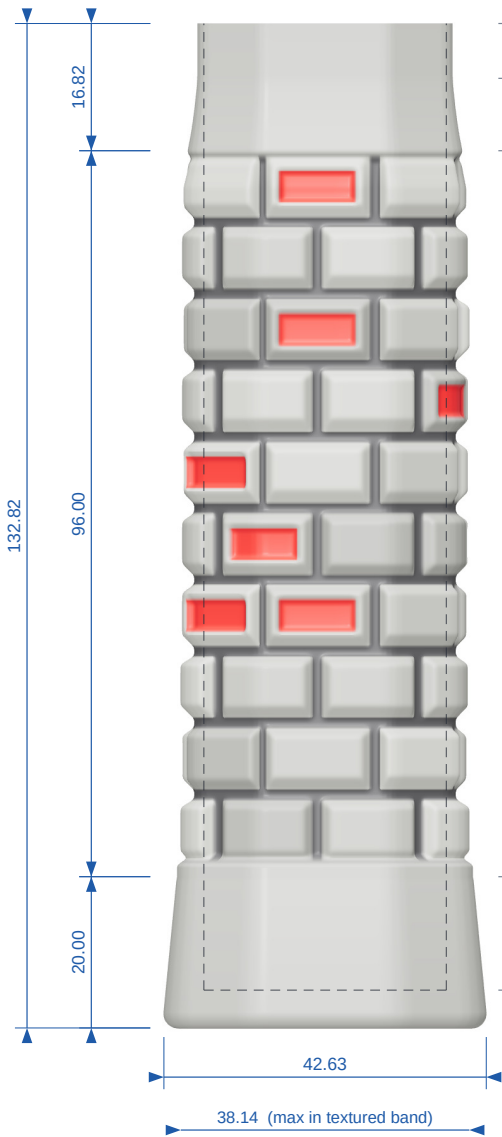
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



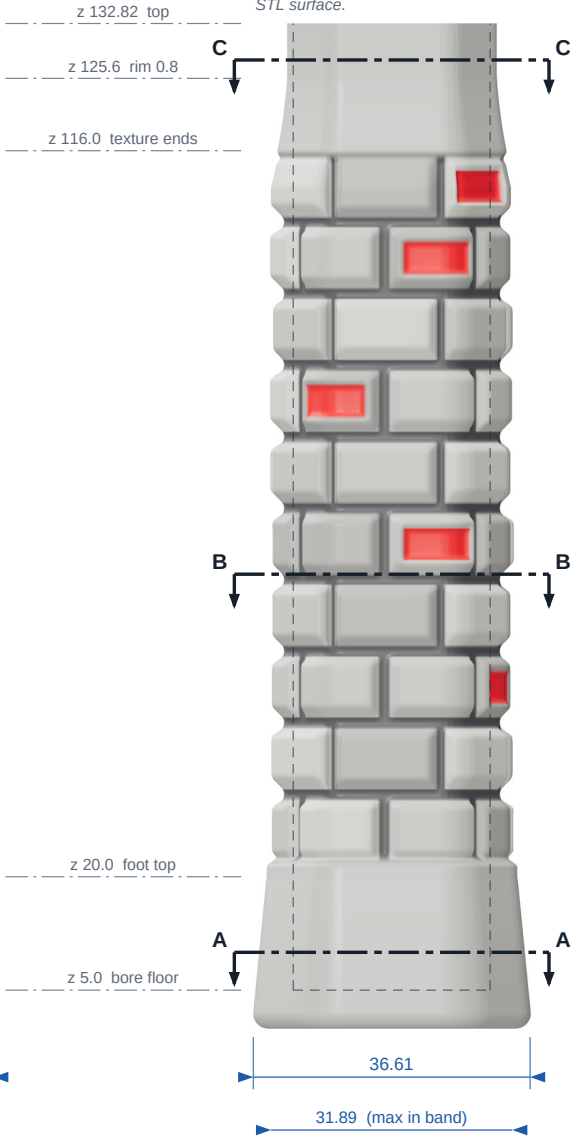
TOP VIEW (from +Z)



FRONT VIEW (from -Y)

KEY DIMENSIONS	
Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	38.14 × 31.89
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.04
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

A, B, C: see sheet 3

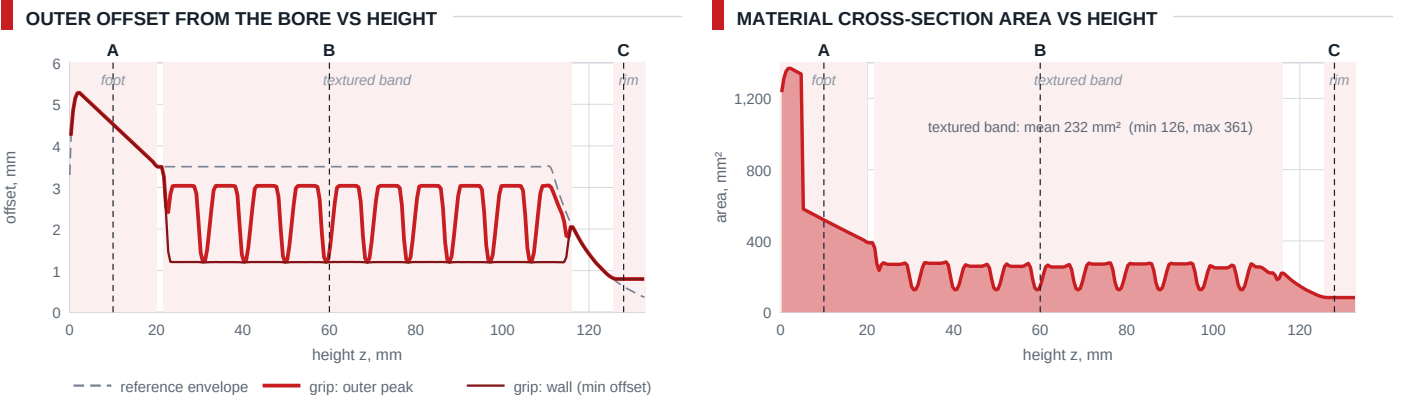
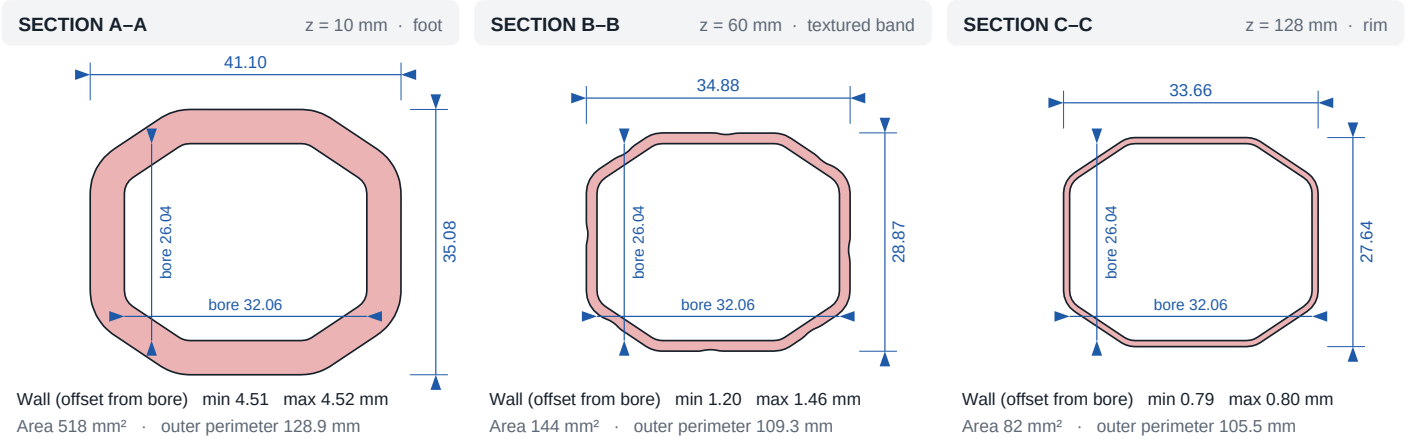
SCALE 1 : 1



PART CELLULAR MOD Pickleball paddle grip sleeve, closed butt, open throat	SCALE 1 : 1	UNITS millimetres	PROJECTION
	MATERIAL PLA / PETG / TPU	INFILL 100 %	LAYER HEIGHT 0.20 mm
GENERAL NOTES 1. Views are true size: print this sheet at 100 %, do not fit to page. 2. Dimensions are measured from the STL, not nominal. 3. Bore, floor and outer envelope match the reference PickleballGrip_1.3mf. 4. Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).	MASS (PLA) 47.7 g	VOLUME 38.47 cm³	SURFACE AREA 323 cm²
	REVISION 2	DATE 2026-09-30	SHEET 2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	38.14 × 31.89 mm
Circumference in band	108.1 – 118.4 mm
Volume / surface area	38.47 cm³ / 322.5 cm²

ZONES BY HEIGHT

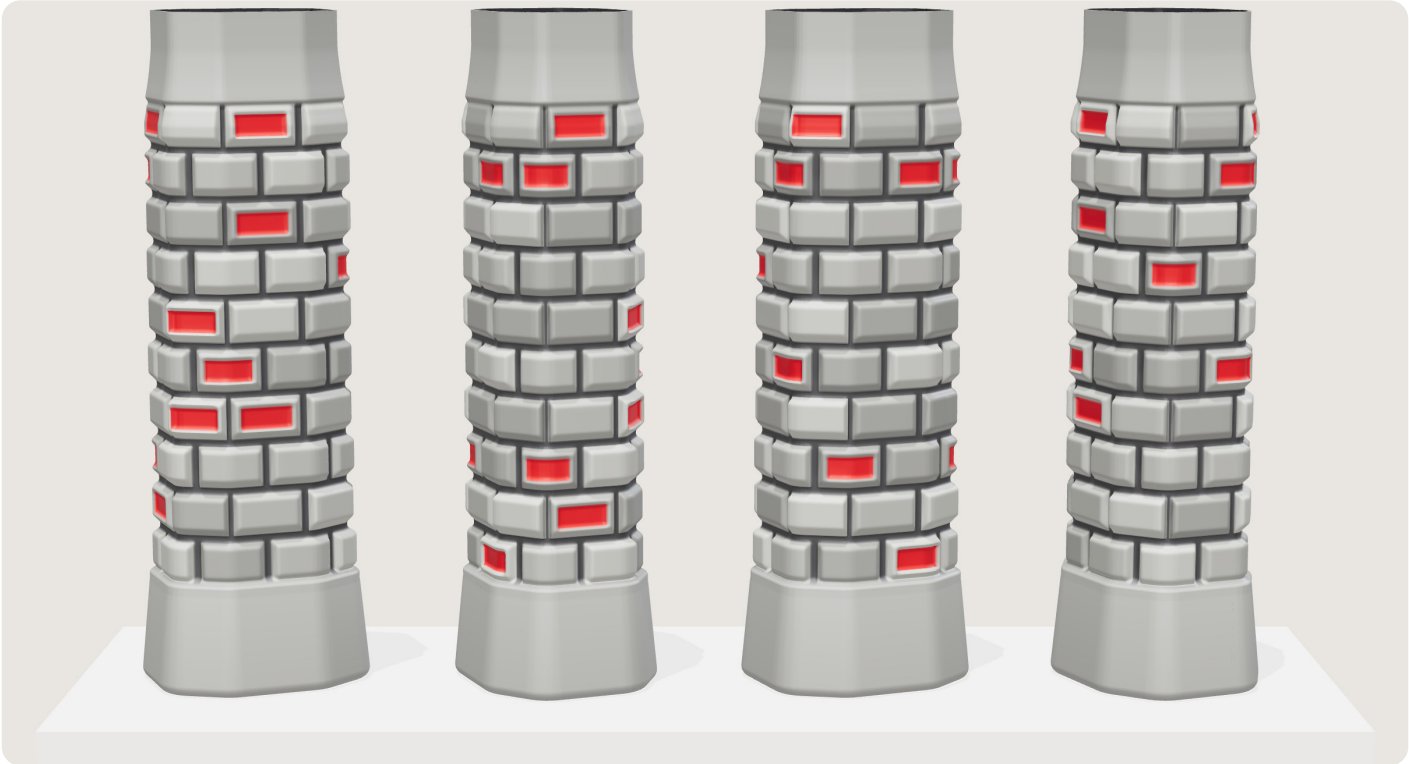
ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.20 mm
Rim wall (design / measured min)			0.80 / 0.80 mm
Peak texture height above bore			3.04 mm

OUTER DIMENSIONS BY HEIGHT

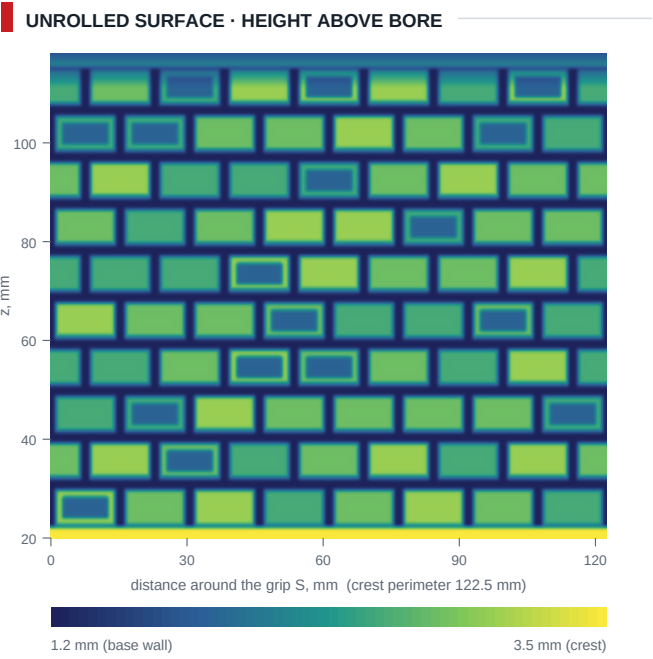
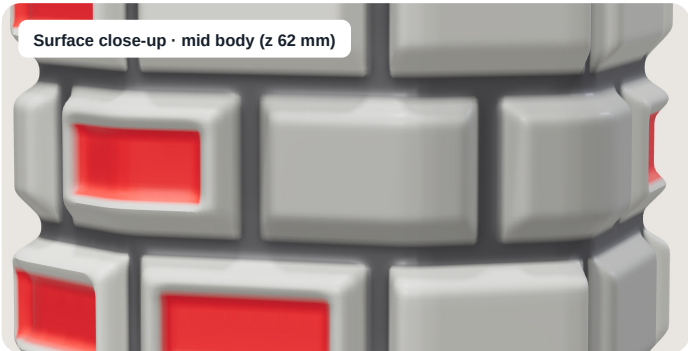
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	35.66	29.57	111.5	176	1.20	1.80
45.0	37.91	31.43	117.0	258	1.21	3.04
60.0	34.96	28.95	109.6	148	1.20	1.46
75.0	37.46	31.66	117.7	271	1.21	3.04
90.0	37.54	31.46	117.3	271	1.21	2.91
105.0	36.98	30.96	115.5	237	1.21	2.64
116.0	36.16	30.14	113.4	219	2.04	2.05
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



DESIGN PARAMETERS	
Rows × cells	10 × 8 (staggered)
Row pitch	9.45 mm
Cell size	15.31 × 8.05 mm
Gap between cells	1.4 mm
Cell heights	2.58 / 2.81 / 3.04 mm
Pocket floor	1.84 mm above bore
Groove floor	1.20 mm (base wall)
PRINT & VERIFICATION	
Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

03

TESSEL-BLOCK

Faceted triangular pyramids in staggered rows for enhanced traction.

46.4 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	38.9 × 32.9 mm
Peak texture height	3.44 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.45 cm ³ / 320 cm ²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm ³		46.4 g
PETG	1.27 g/cm ³		47.6 g
TPU	1.21 g/cm ³		45.3 g
ABS	1.04 g/cm ³		38.9 g

50 g limit

COLOURWAY

#848482	#E8742A	#387AC4

3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.6 m (100 % flow)



3/4 view · rendered from the STL, colours are preview only

DESIGN INTENT

A tessellation of triangular pyramids. Traction comes from many small angled edges rather than one deep texture, and up-pointing pyramids stand taller than down-pointing ones, giving a two-level, gem-like surface.

Facet slopes stay below 45 degrees; rim rows are flat grey bands in the colourway.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 11 rows × 24 triangles (12 up + 12 down per row)
- Sharp V-grooves between facets
- Up-pyramids 3.44 mm, down-pyramids 3.04 mm above the bore

FILES

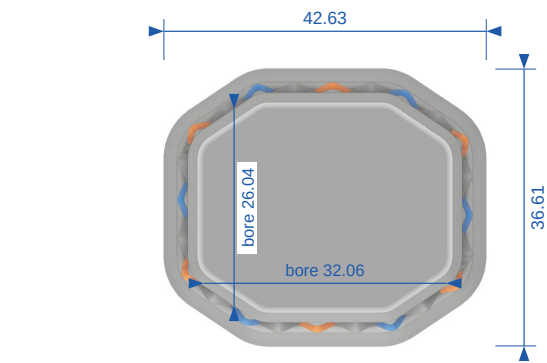
stl/03_tessel_block.stl	19.9 MB · 398,860 triangles
3mf/03_tessel_block.3mf	4.6 MB · colour

✓ FITMENT VERIFIED

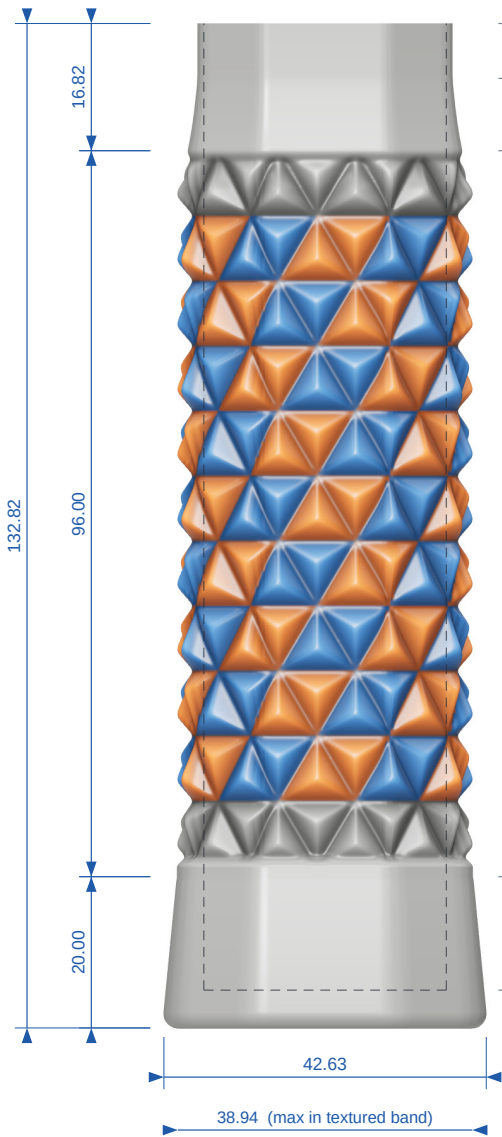
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



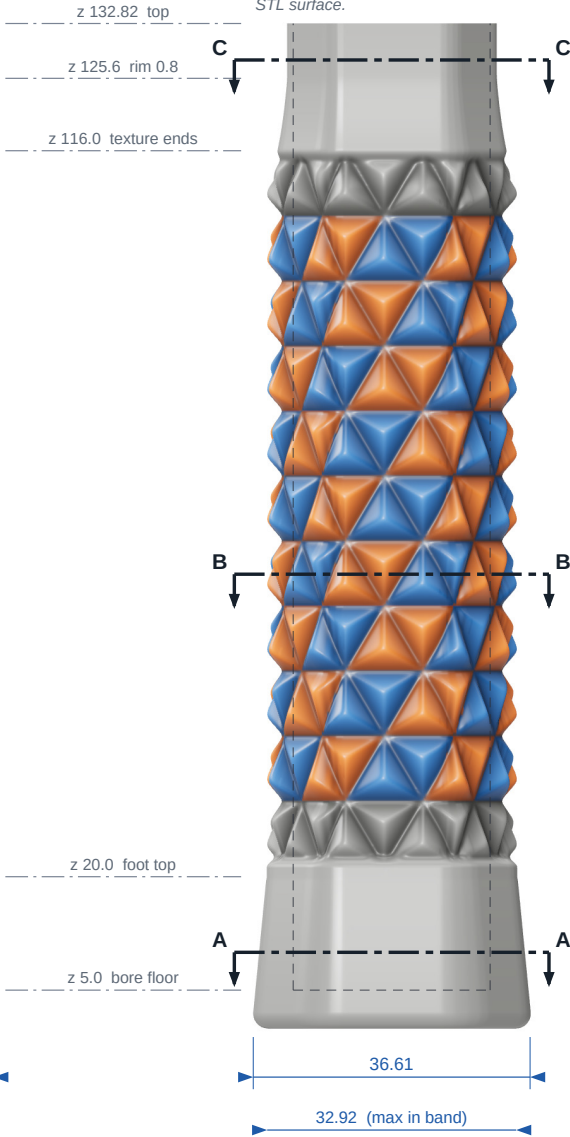
TOP VIEW (from +Z)



FRONT VIEW (from -Y)

KEY DIMENSIONS	
Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	38.94 × 32.92
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.44
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

A, B, C: see sheet 3

SCALE 1 : 1



PART

TESSEL-BLOCK

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

MASS (PLA)

46.4 g

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

37.45 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

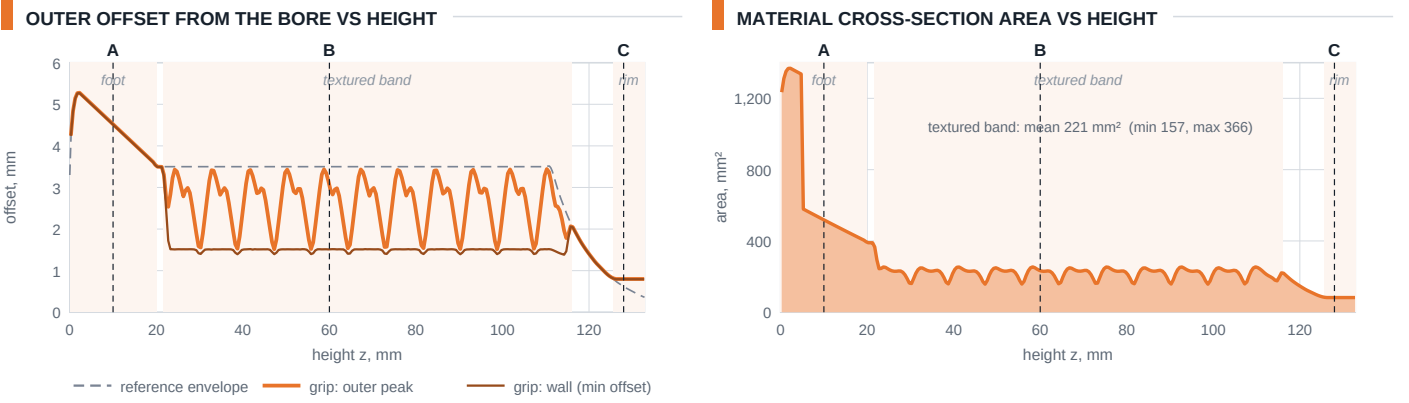
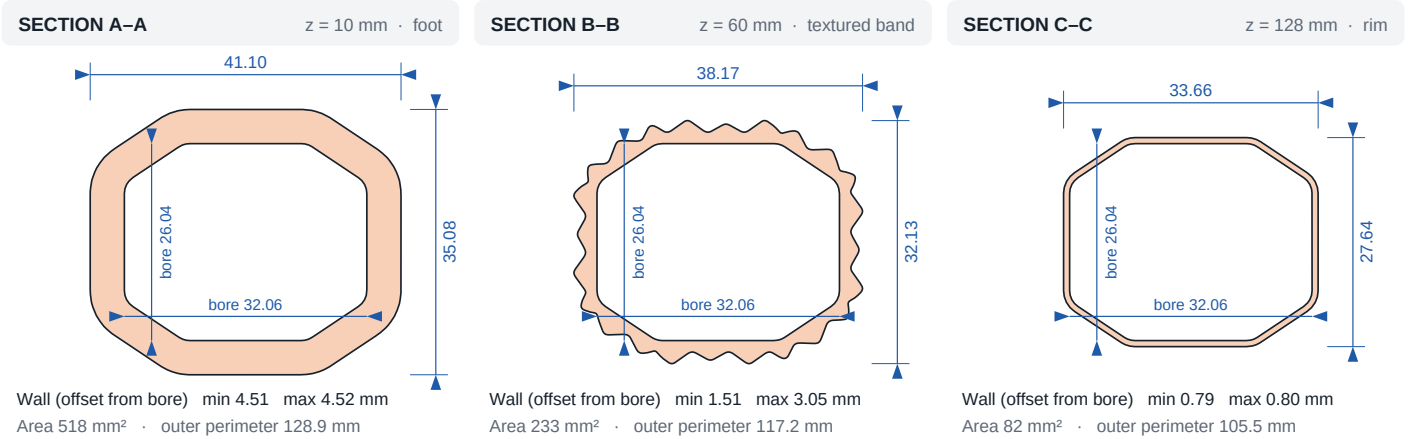
320 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	38.94 × 32.92 mm
Circumference in band	109.9 – 120.8 mm
Volume / surface area	37.45 cm ³ / 320.4 cm ²

ZONES BY HEIGHT

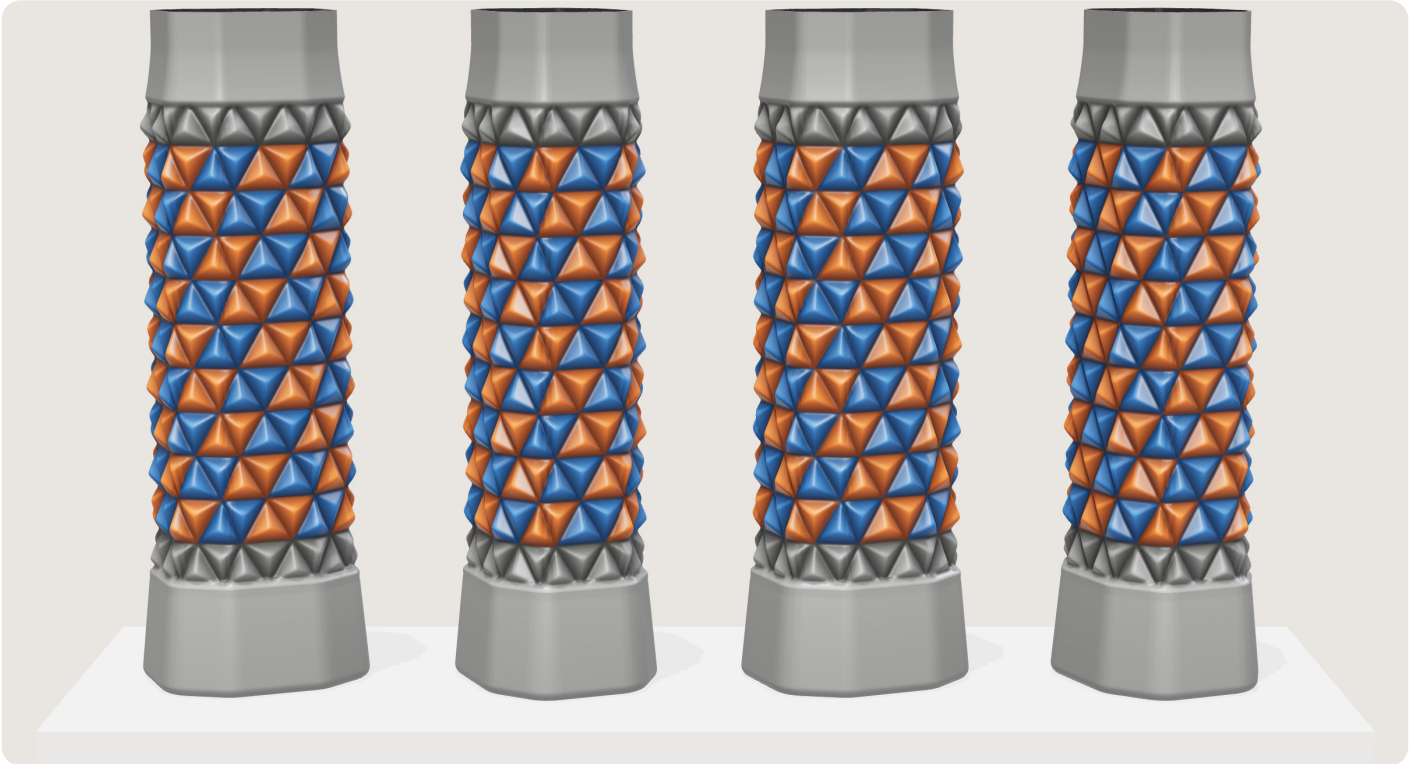
ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.39 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			3.44 mm

OUTER DIMENSIONS BY HEIGHT

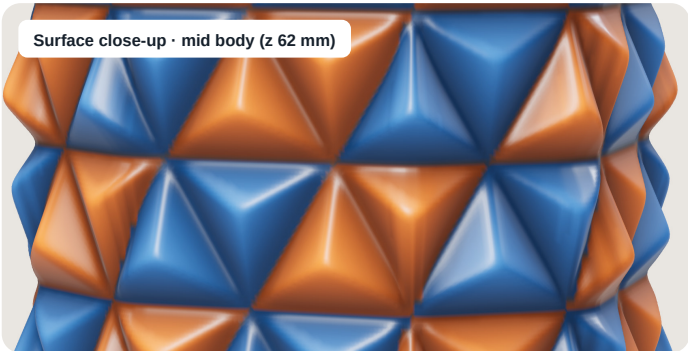
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	35.21	29.18	110.3	162	1.41	1.57
45.0	37.70	31.68	117.3	230	1.51	2.82
60.0	38.17	32.13	117.3	233	1.51	3.05
75.0	38.06	32.03	116.9	249	1.51	3.00
90.0	35.24	29.22	110.3	164	1.42	1.59
105.0	37.81	31.79	115.9	234	1.51	2.87
116.0	36.18	30.16	113.4	220	2.05	2.06
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

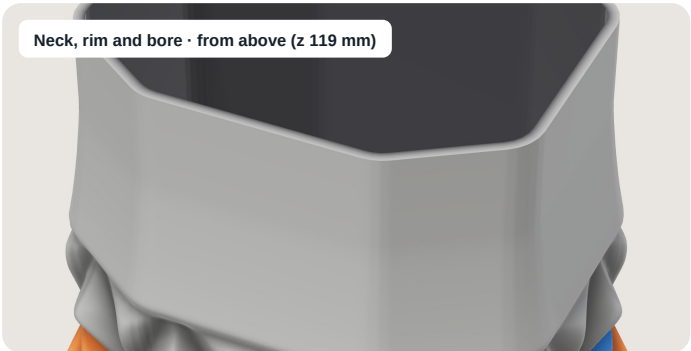
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only

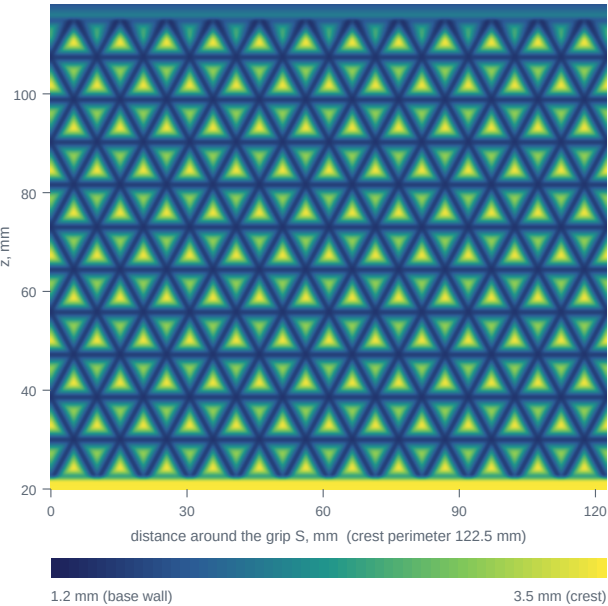


Surface close-up · mid body (z 62 mm)



Neck, rim and bore · from above (z 119 mm)

UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Rows	11
Triangles per row	24 (12 up + 12 down)
Triangle base	10.21 mm
Row height	8.59 mm
Up-pyramid apex	3.44 mm above bore
Down-pyramid apex	3.04 mm above bore
Groove floor	about 1.4 mm above bore

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

04

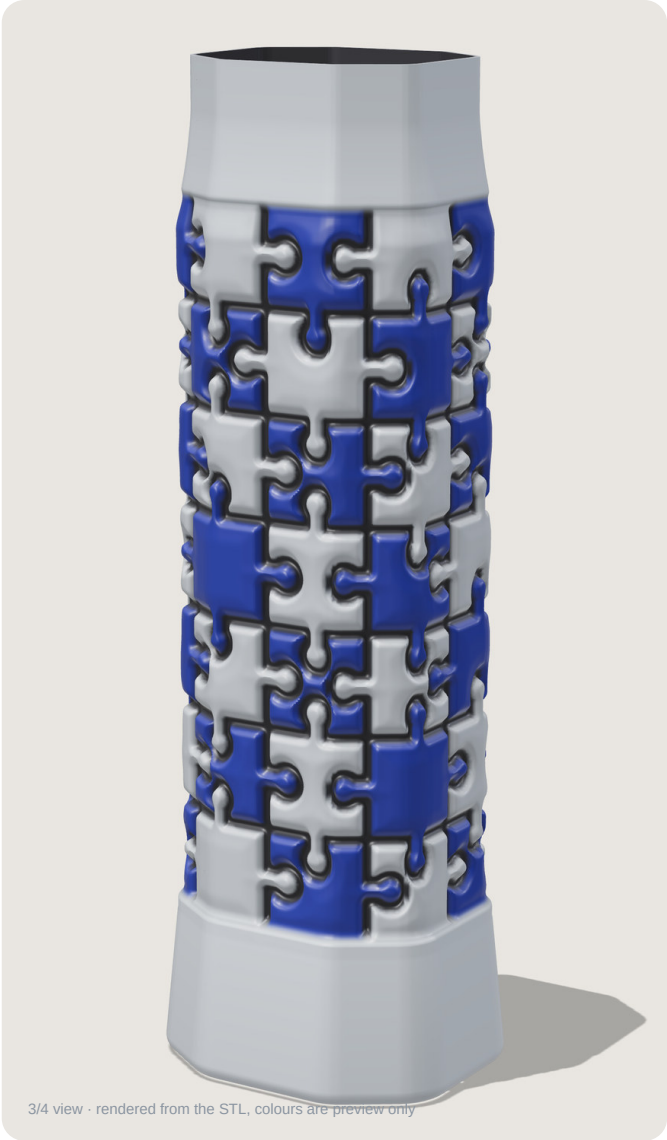
LOGIC-GRIP

Seamlessly interlocking ergonomic puzzle design.

48.0 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	37.7 × 31.7 mm
Peak texture height	2.81 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	38.73 cm³ / 330 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		48.0 g
PETG	1.27 g/cm³		49.2 g
TPU	1.21 g/cm³		46.9 g
ABS	1.04 g/cm³		40.3 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 16.1 m (100 % flow)

DESIGN INTENT

Fifty-six jigsaw pieces tile the grip in alternating colours. Each piece is a shallow pillow separated by fine grooves and locked to its neighbours by mushroom knobs; knob directions are randomised so no two rows repeat.

Grooves are 0.84 mm wide, about two nozzle widths, and 1.6 mm deep.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 8 × 7 = 56 pieces in a checkerboard colourway
- Mushroom knobs: 2.9 mm neck, 4.2 mm head, 5.1 mm reach
- 0.84 mm grooves cut down to the base wall

FILES

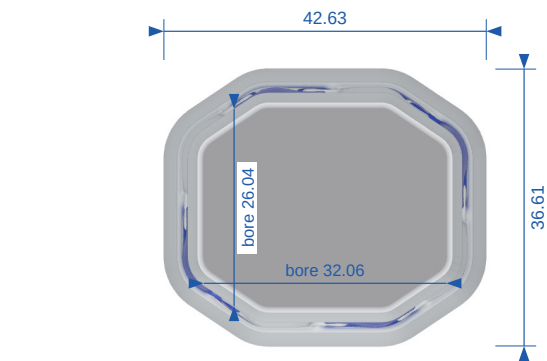
stl/04_logic_grip.stl	19.8 MB · 396,900 triangles
3mf/04_logic_grip.3mf	4.6 MB · colour

✓ FITMENT VERIFIED

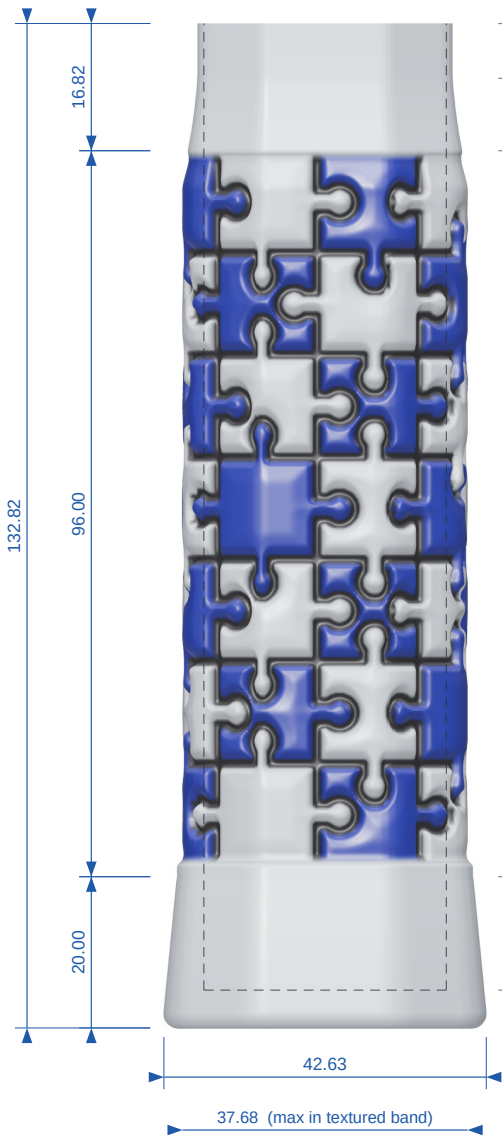
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

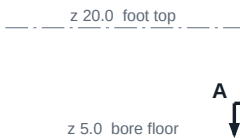
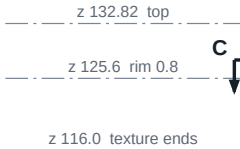
✓ 3MF VALIDATED



TOP VIEW (from +Z)



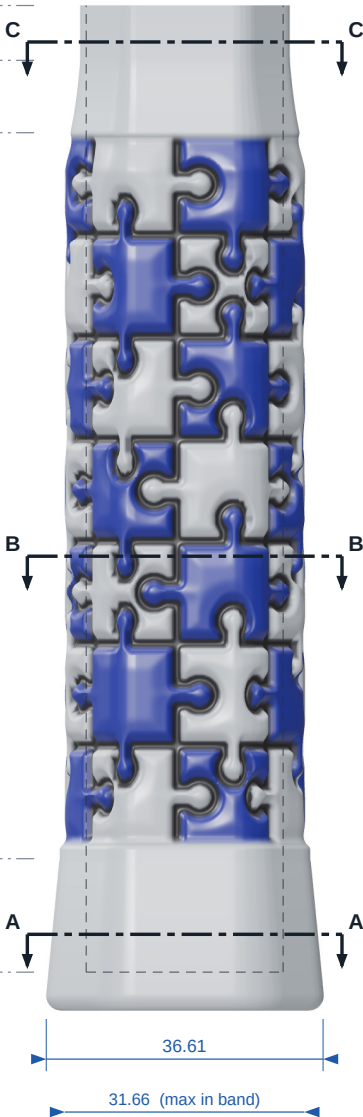
FRONT VIEW (from -Y)



KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	37.68 × 31.66
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	2.81
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

A, B, C: see sheet 3

SCALE 1 : 1



PART

LOGIC-GRIP

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

38.73 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

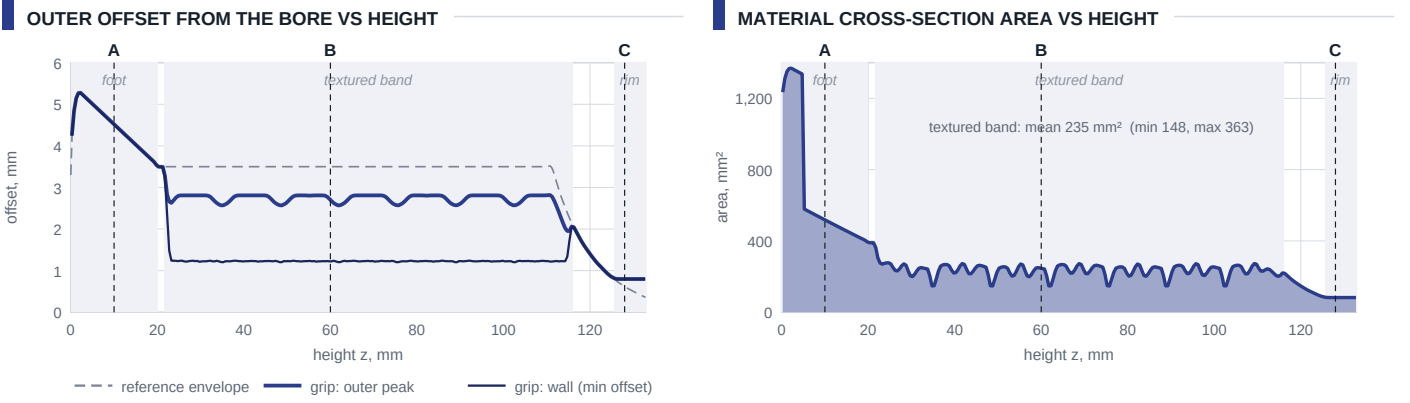
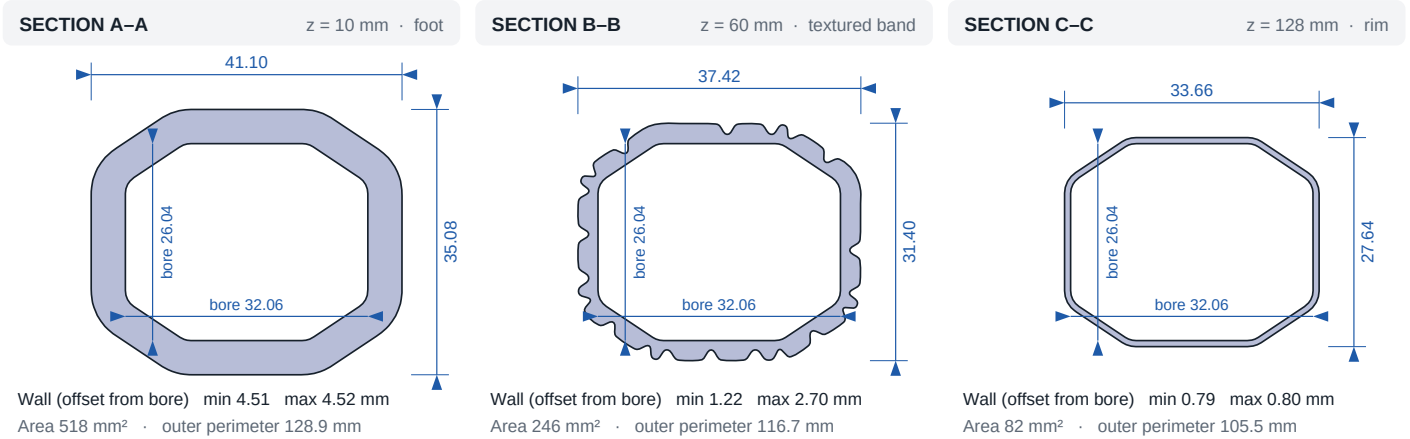
330 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	37.68 × 31.66 mm
Circumference in band	114.5 – 117.7 mm
Volume / surface area	38.73 cm ³ / 329.9 cm ²

ZONES BY HEIGHT

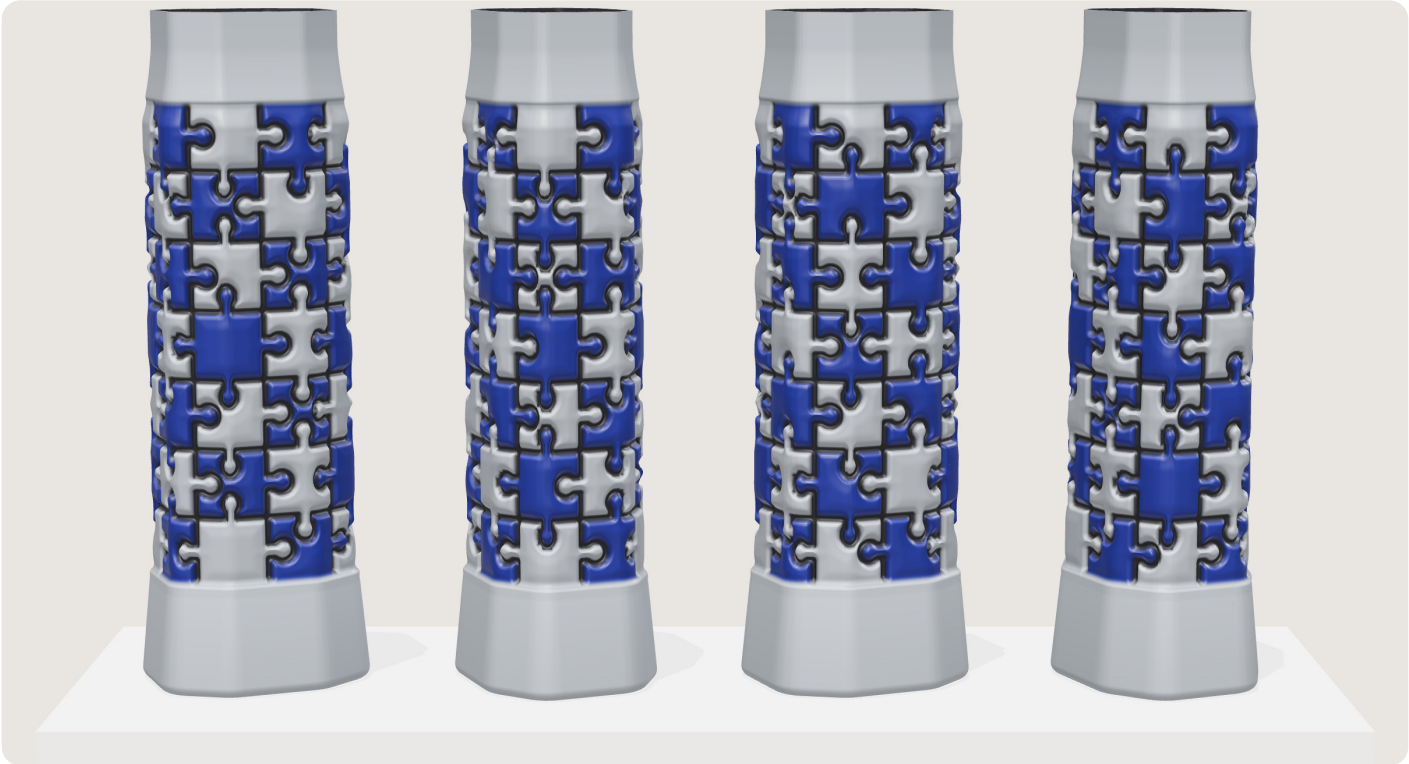
ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.20 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			2.81 mm

OUTER DIMENSIONS BY HEIGHT

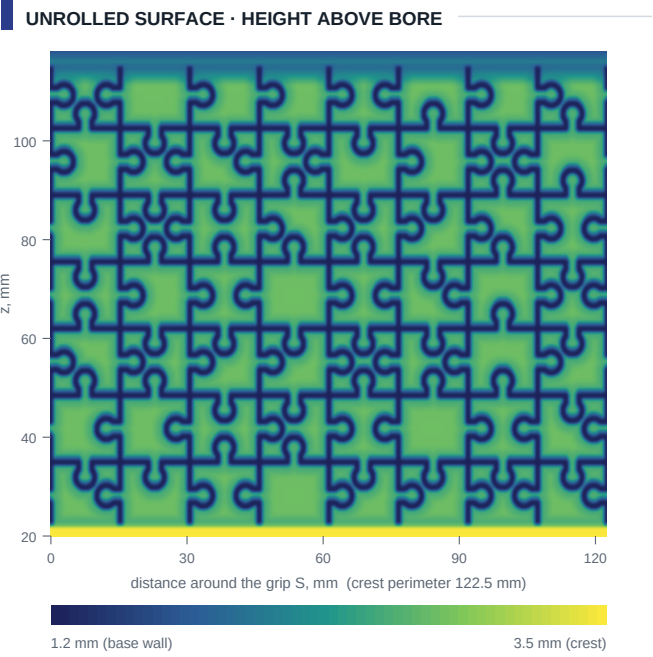
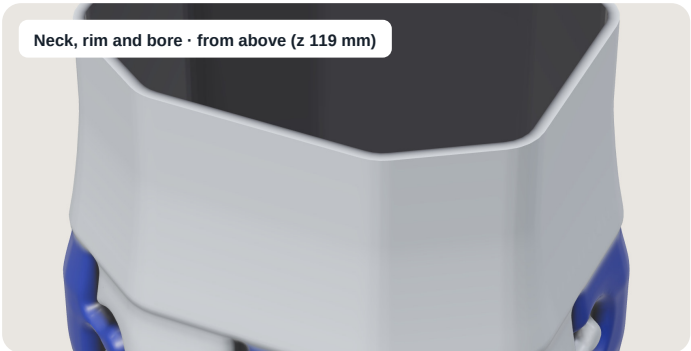
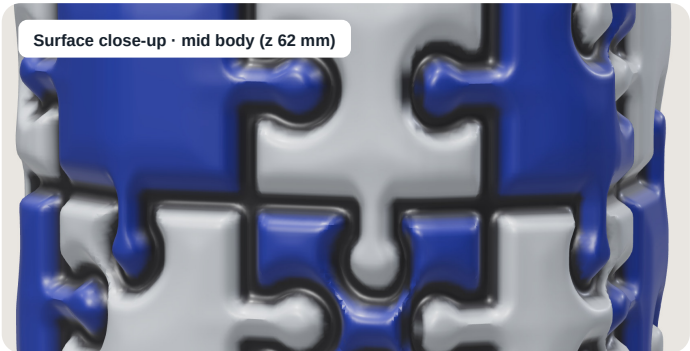
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	37.57	31.60	115.1	202	1.22	2.81
45.0	37.59	31.65	117.1	254	1.23	2.81
60.0	37.42	31.40	116.7	247	1.22	2.70
75.0	37.23	31.19	114.9	179	1.21	2.58
90.0	37.24	31.27	115.3	208	1.23	2.61
105.0	37.56	31.50	117.2	262	1.23	2.75
116.0	36.17	30.15	113.4	220	2.04	2.05
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



DESIGN PARAMETERS	
Pieces	8 × 7 = 56
Piece size	15.31 × 13.50 mm
Knob neck / head	2.9 mm / dia 4.2 mm
Knob reach	5.1 mm
Groove width	0.84 mm
Groove floor	1.20 mm (base wall)
Piece top height	up to 2.81 mm above bore
PRINT & VERIFICATION	
Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

05

NEURO-TREAD

Bio-inspired raised neural network over a dimpled adaptive base.

46.5 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	39.1 × 33.0 mm
Peak texture height	3.50 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.49 cm³ / 330 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		46.5 g
PETG	1.27 g/cm³		47.6 g
TPU	1.21 g/cm³		45.4 g
ABS	1.04 g/cm³		39.0 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.6 m (100 % flow)

DESIGN INTENT

A network of 48 neurons joined by curved axon ridges rises to the full 3.5 mm crest, standing proud of a field of small bumps. The ridges guide the fingers while the bump field provides fine grip.

Ridges and bumps are gentle slopes; the network is printed in the same material as the base.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 48 somas (8 × 6) joined by curved axons about 1.5 mm wide
- Bump field on a 2.66 mm hex pitch
- Network and somas reach the full 3.5 mm crest

FILES

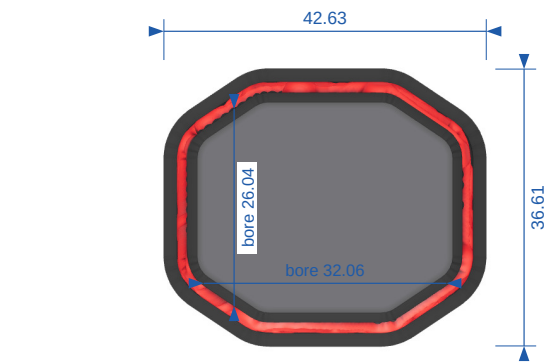
stl/05_neuro_tread.stl	19.9 MB · 398,860 triangles
3mf/05_neuro_tread.3mf	4.8 MB · colour

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



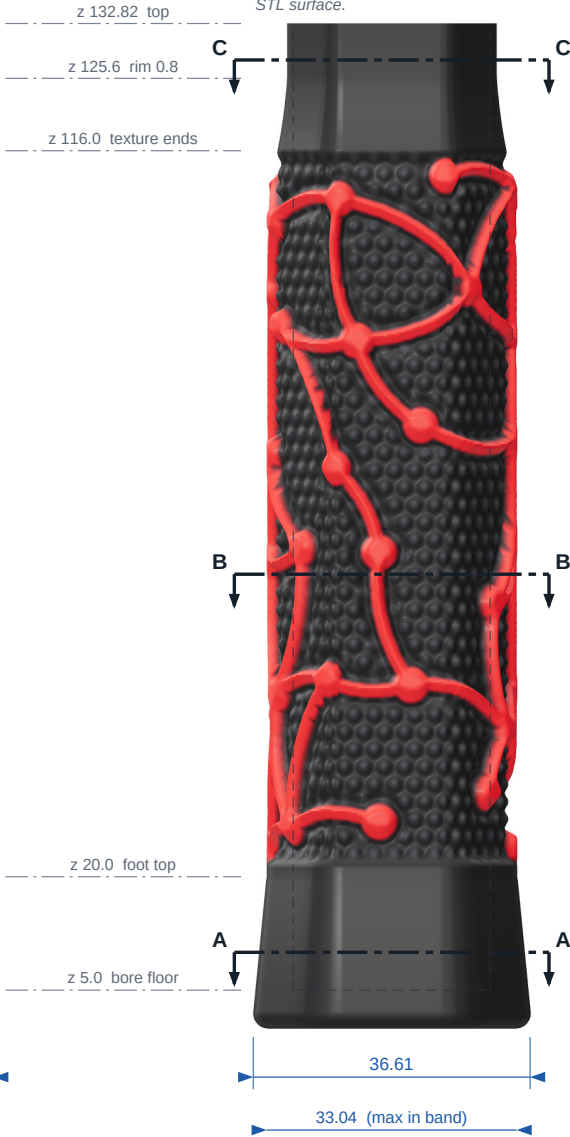
TOP VIEW (from +Z)



FRONT VIEW (from -Y)

KEY DIMENSIONS	
Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	39.06 × 33.04
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.50
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

A, B, C: see sheet 3

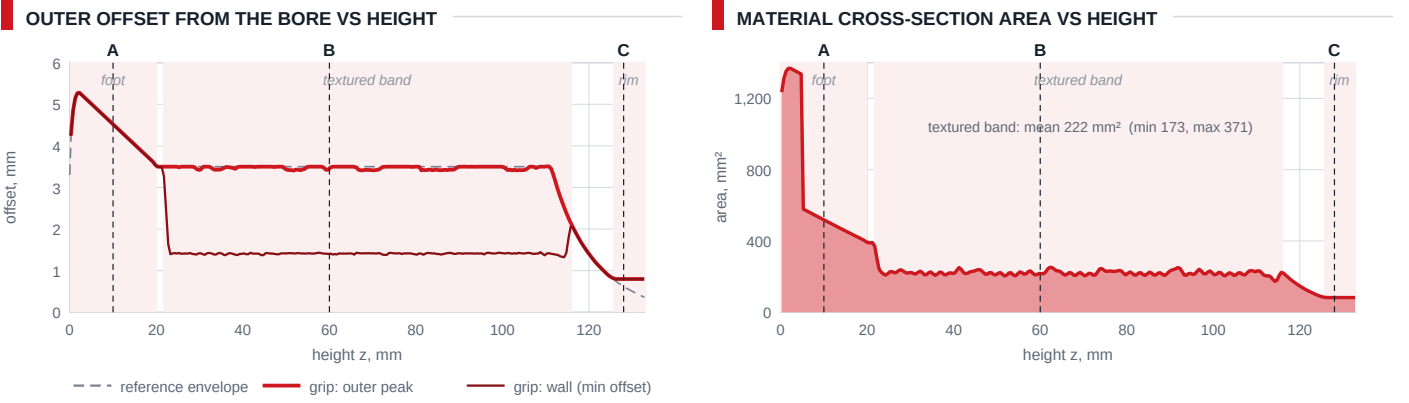
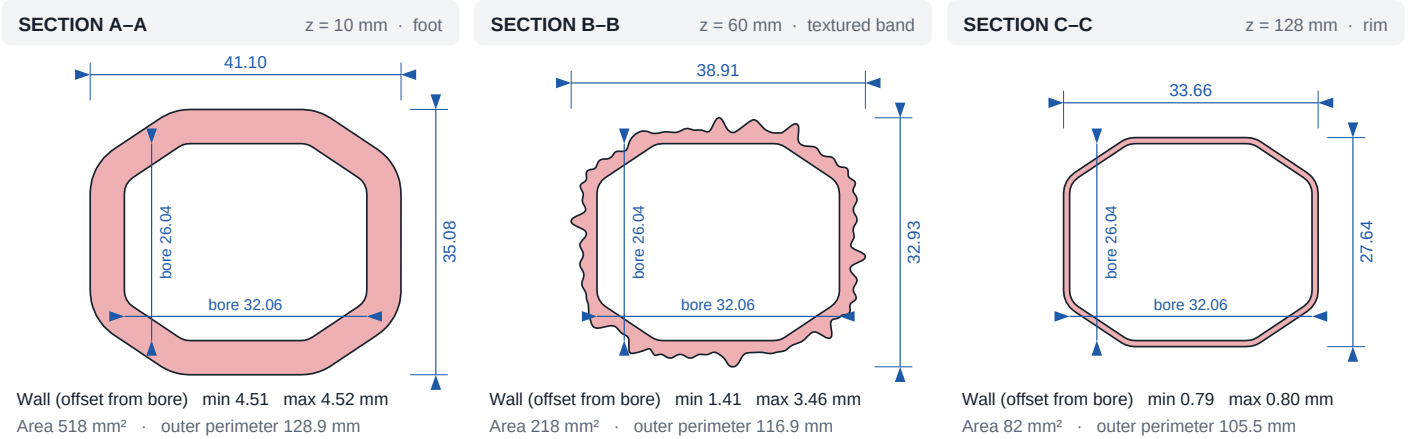
SCALE 1 : 1



PART NEURO-TREAD Pickleball paddle grip sleeve, closed butt, open throat	SCALE 1 : 1	UNITS millimetres	PROJECTION
	MATERIAL PLA / PETG / TPU	INFILL 100 %	LAYER HEIGHT 0.20 mm
GENERAL NOTES 1. Views are true size: print this sheet at 100 %, do not fit to page. 2. Dimensions are measured from the STL, not nominal. 3. Bore, floor and outer envelope match the reference PickleballGrip_1.3mf. 4. Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).	MASS (PLA) 46.5 g	VOLUME 37.49 cm³	SURFACE AREA 330 cm²
	REVISION 2	DATE 2026-09-30	SHEET 2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	39.06 × 33.04 mm
Circumference in band	114.1 – 119.7 mm
Volume / surface area	37.49 cm ³ / 329.7 cm ²

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.37 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			3.50 mm

OUTER DIMENSIONS BY HEIGHT

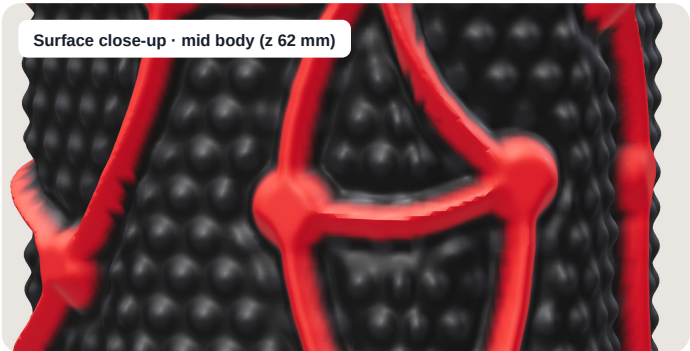
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	37.77	31.75	116.8	222	1.42	3.43
45.0	38.99	32.88	117.1	234	1.40	3.50
60.0	38.91	32.93	116.9	217	1.41	3.46
75.0	38.94	32.96	119.3	232	1.42	3.50
90.0	38.99	32.86	117.2	231	1.42	3.50
105.0	38.74	32.79	116.7	208	1.41	3.42
116.0	36.24	30.19	113.5	221	2.05	2.10
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only

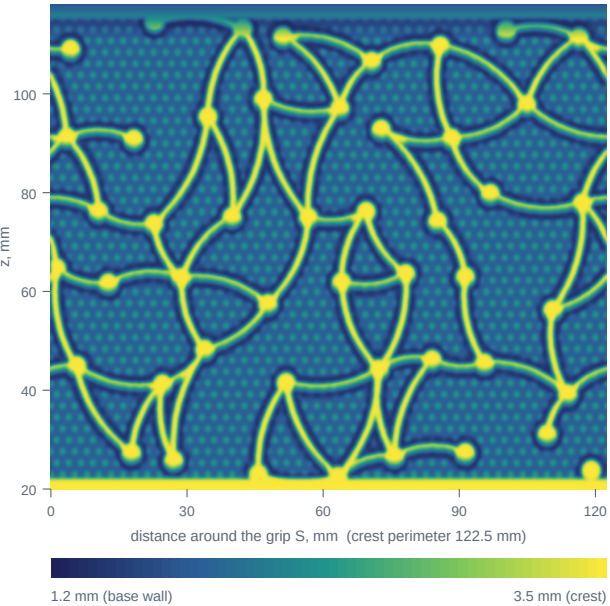


Surface close-up · mid body (z 62 mm)



Neck, rim and bore · from above (z 119 mm)

UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Neurons (somas)	48 (8 × 6)
Axon ridge width	about 1.5 mm
Soma diameter	about 4.2 mm
Network height	3.50 mm above bore
Bump pitch	2.66 mm (hex)
Bump field height	1.4 – 2.4 mm above bore
Ridge / bump colours	red / black

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

06

CARBON MATRIX

Continuous 2/2 twill weave with selective reinforcement patches.

48.1 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	37.5 × 31.4 mm
Peak texture height	2.73 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	38.77 cm³ / 307 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		48.1 g
PETG	1.27 g/cm³		49.2 g
TPU	1.21 g/cm³		46.9 g
ABS	1.04 g/cm³		40.3 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 16.1 m (100 % flow)

DESIGN INTENT

A woven-carbon look built as real geometry: every tow rises over and dives under its neighbours in a 2/2 twill. Smooth, slightly raised patches suggest selective reinforcement where the palm loads the handle.

The weave is the finest texture in the set (0.4 mm relief on a 2 mm pitch).

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 60 tows around, 2.04 mm pitch, 2/2 twill
- Weave relief about 0.4 mm; every tow is a rounded ridge
- Smooth reinforcement patches stand about 0.5 mm above the weave

FILES

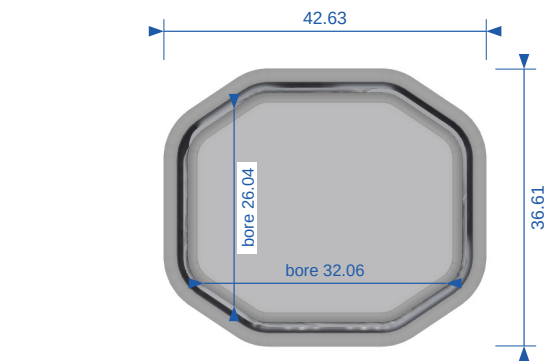
stl/06_carbon_matrix.stl	19.9 MB · 398,860 triangles
3mf/06_carbon_matrix.3mf	4.7 MB · colour

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



TOP VIEW (from +Z)

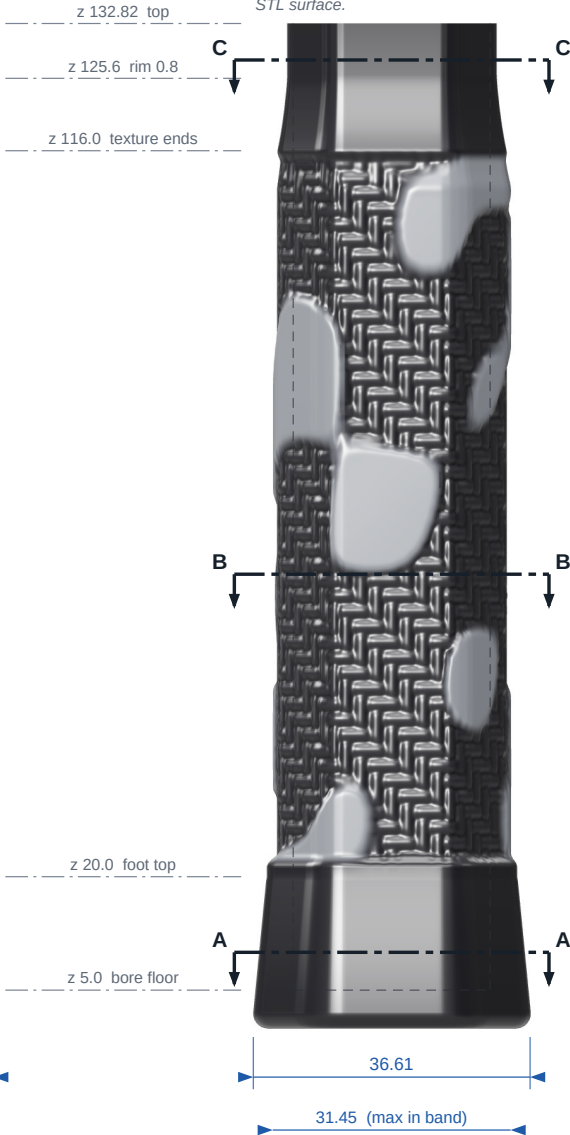


FRONT VIEW (from -Y)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	37.46 × 31.45
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	2.73
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

A, B, C: see sheet 3

SCALE 1 : 1



PART

CARBON MATRIX

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

MASS (PLA)

48.1 g

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

38.77 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

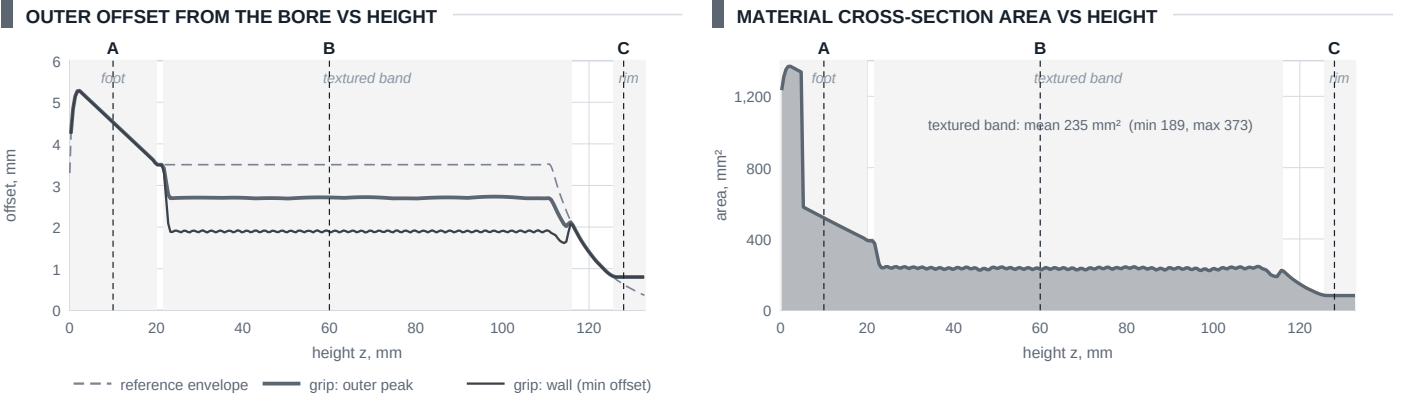
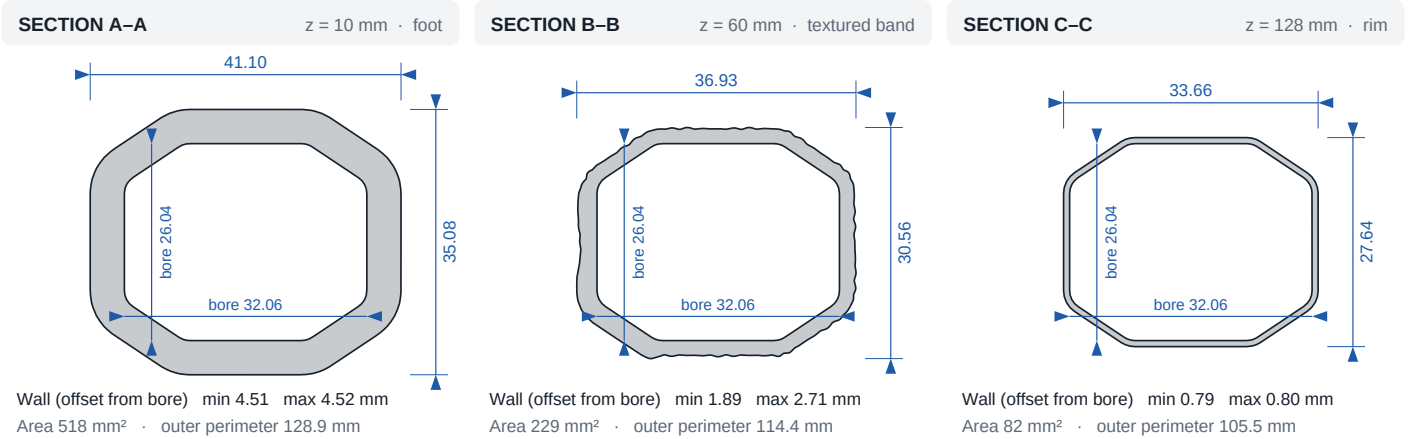
307 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	37.46 × 31.45 mm
Circumference in band	113.7 – 115.2 mm
Volume / surface area	38.77 cm³ / 306.6 cm²

ZONES BY HEIGHT

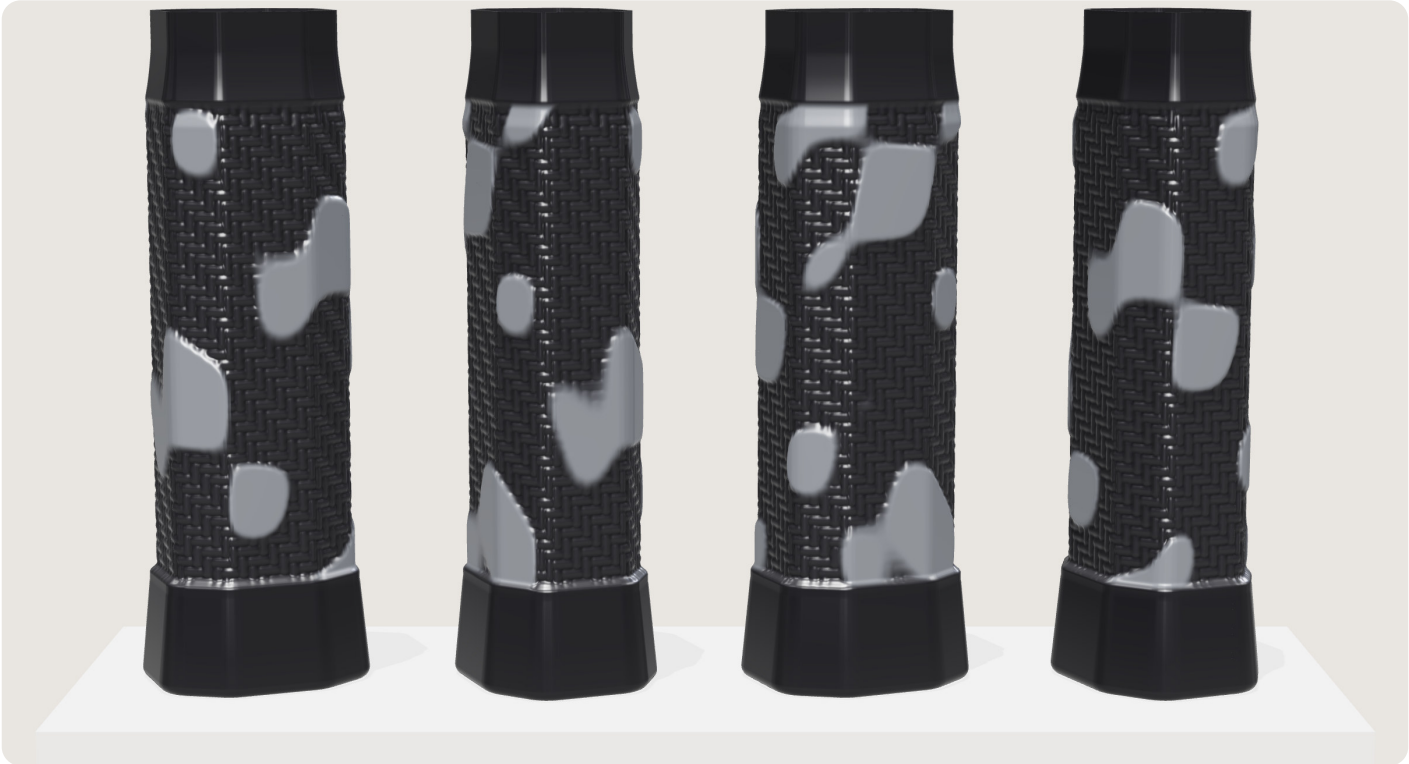
ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.87 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			2.73 mm

OUTER DIMENSIONS BY HEIGHT

Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	36.88	30.88	114.8	237	1.89	2.71
45.0	37.26	31.01	114.5	237	1.91	2.70
60.0	36.96	30.56	114.5	230	1.89	2.71
75.0	36.88	30.87	114.5	237	1.89	2.69
90.0	36.78	31.08	114.9	238	1.91	2.71
105.0	36.89	30.87	114.7	236	1.88	2.70
116.0	36.21	30.20	113.5	221	2.06	2.08
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



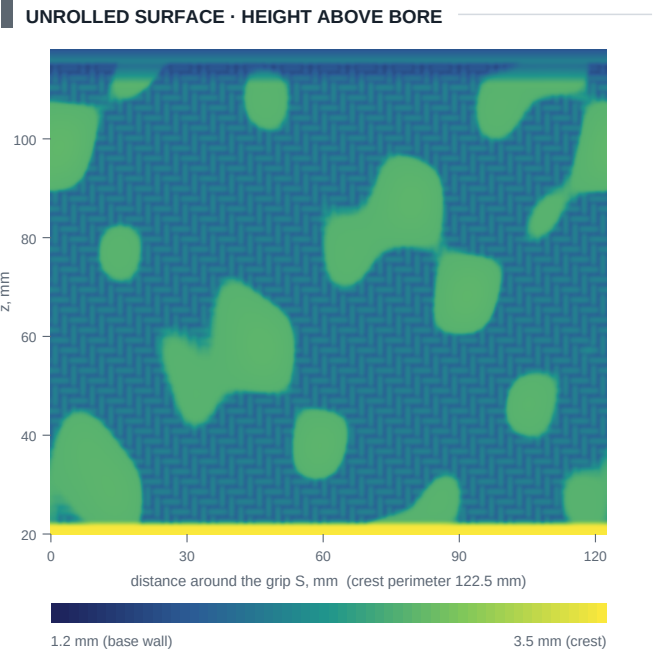
Four views, 90° apart · rendered from the STL, colours are preview only



Surface close-up · mid body (z 62 mm)



Neck, rim and bore · from above (z 119 mm)



DESIGN PARAMETERS	
Tows around	60 (2/2 twill)
Tow pitch	2.04 mm
Weave heights	1.7 – 2.2 mm above bore
Patch heights	2.3 – 2.7 mm above bore
Patch coverage	about 23% of the band
Weave / patch colours	carbon / grey
Textured band	94.5 mm tall
PRINT & VERIFICATION	
Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

07

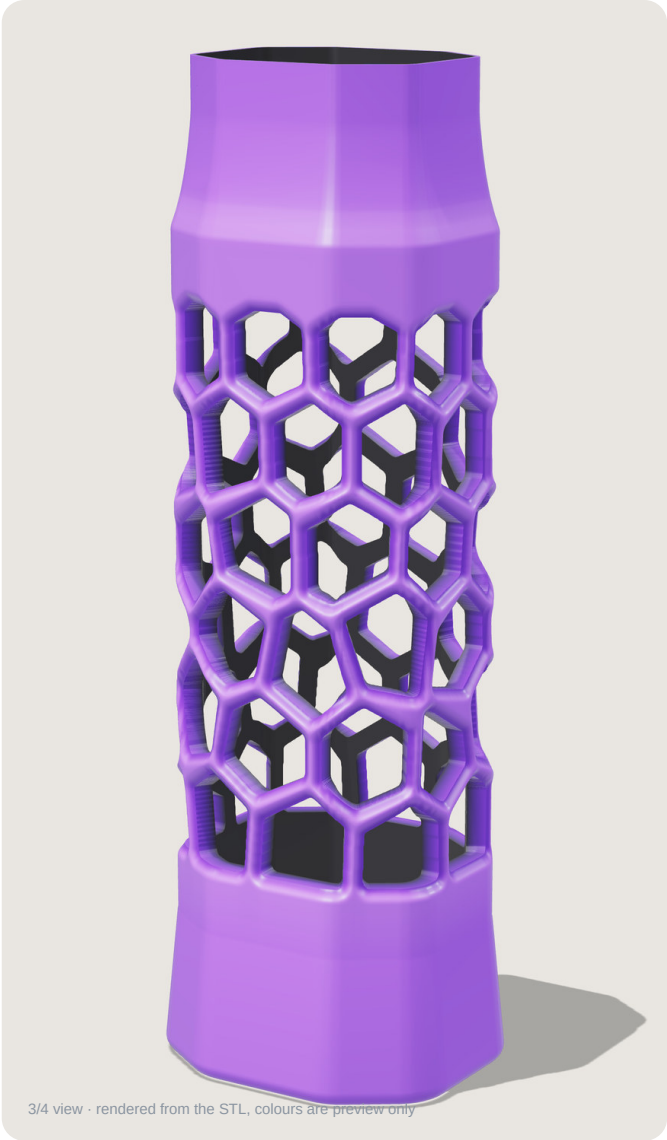
VORONOI-CORE

Voronoi rib lattice, windows cut through the wall for weight saving and airflow.

40.2 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	39.1 × 33.0 mm
Peak texture height	3.50 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	32.41 cm³ / 259 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		40.2 g
PETG	1.27 g/cm³		41.2 g
TPU	1.21 g/cm³		39.2 g
ABS	1.04 g/cm³		33.7 g

50 g limit

COLOURWAY



#8A54E4

3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 13.5 m (100 % flow)

DESIGN INTENT

Irregular Voronoi cells become 2.6 mm ribs with rounded crests, and the cells themselves are open windows through the wall – so the grip is ventilated and the lightest of the ten. The lattice is relaxed for even cells but keeps a natural irregularity.

Window roofs are trimmed; the longest bridge is 3.8 mm.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical; 54 windows open the wall
Outer envelope vs reference	+0.003 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	3.8 mm

KEY FEATURES

- 54 through-windows in the band z 27.5 – 103.5 mm
- 2.6 mm ribs standing at the full 3.5 mm crest
- Window roofs trimmed to 60 degrees, longest bridge 3.8 mm

FILES

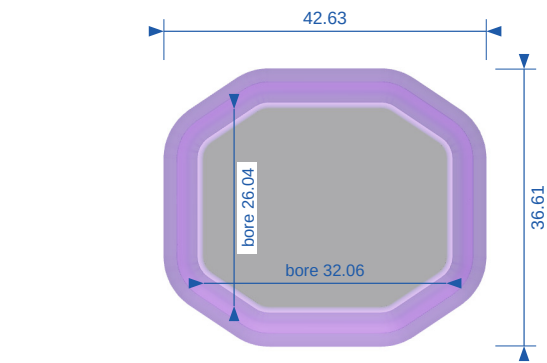
stl/07_voronoi_core.stl	9.7 MB · 194,162 triangles
3mf/07_voronoi_core.3mf	2.3 MB · colour

✓ FITMENT VERIFIED

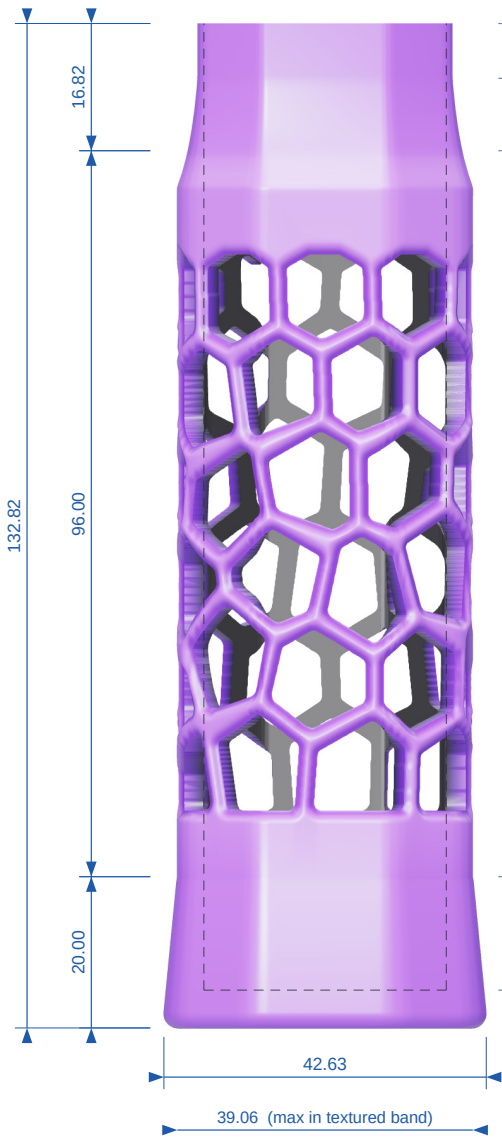
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

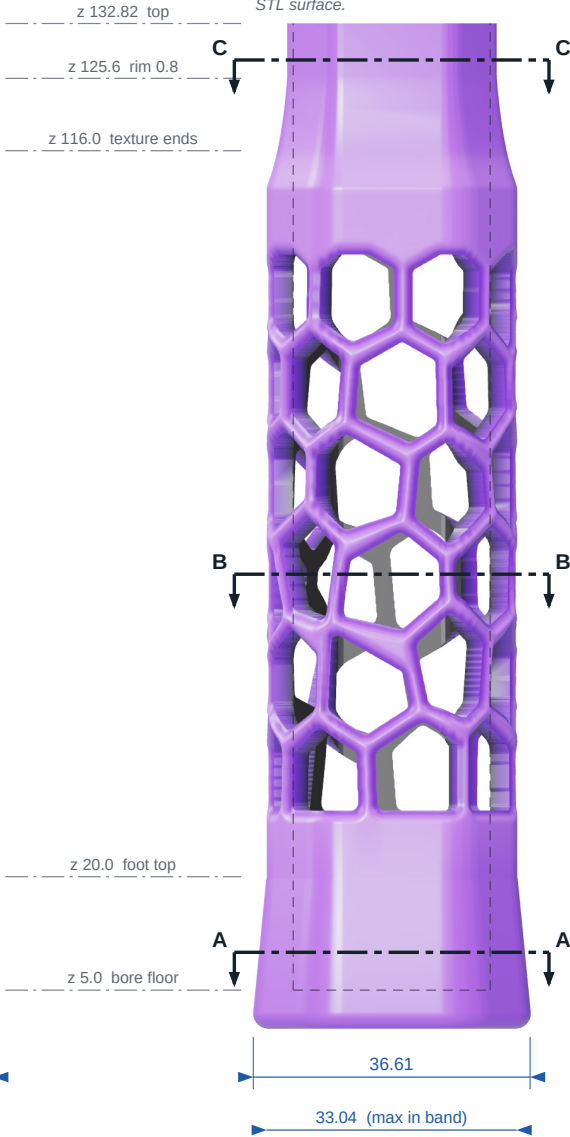
✓ 3MF VALIDATED



TOP VIEW (from +Z)



FRONT VIEW (from -Y)



RIGHT VIEW (from +X)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	39.06 × 33.04
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.50
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.

SCALE 1 : 1



PART

VORONOI-CORE

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

MASS (PLA)

40.2 g

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

32.41 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

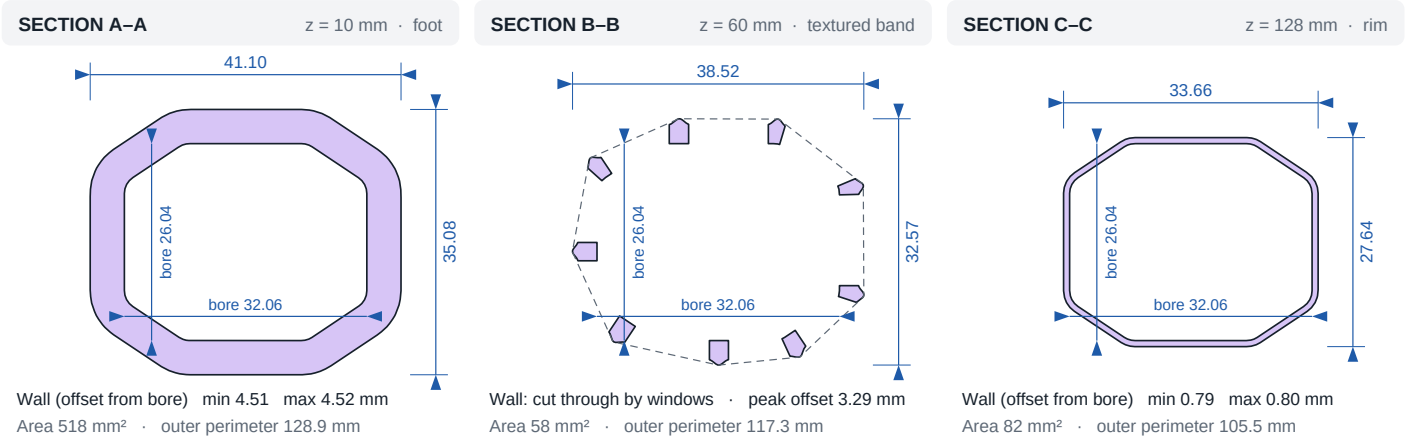
259 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	39.06 × 33.04 mm
Circumference in band	116.3 – 122.5 mm
Volume / surface area	32.41 cm ³ / 259.0 cm ²

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / windows
Rim wall (design / measured min)			0.80 / 0.80 mm
Peak texture height above bore			3.50 mm

OUTER DIMENSIONS BY HEIGHT

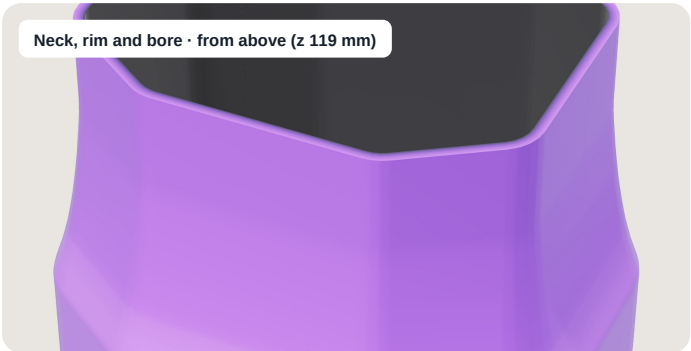
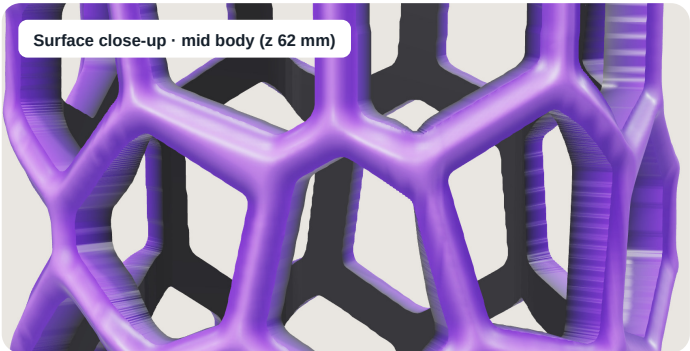
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	38.63	32.65	116.8	62	open	3.31
45.0	38.57	32.62	117.2	59	open	3.30
60.0	38.52	32.57	117.3	58	open	3.29
75.0	38.81	32.61	117.5	88	open	3.50
90.0	38.91	32.86	120.5	212	open	3.46
105.0	39.06	33.04	122.5	390	3.49	3.50
116.0	36.27	30.24	113.7	225	2.10	2.10
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

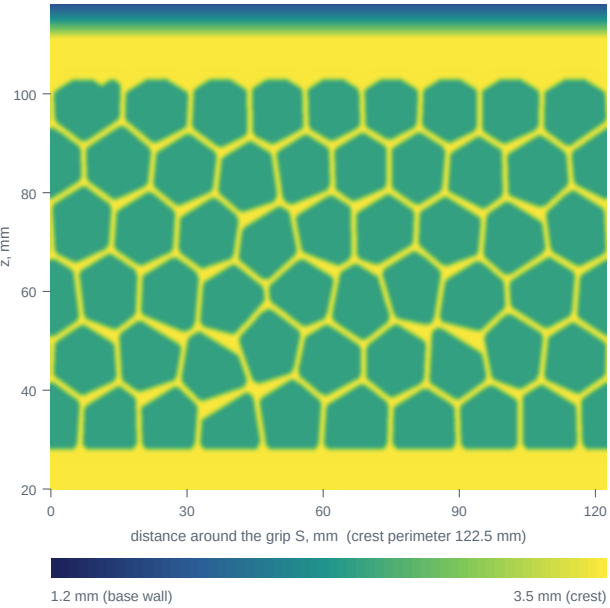
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Windows	54, through the wall
Rib width	2.6 mm
Rib height	3.50 mm above bore
Windowed band	z 27.5 – 103.5 mm (76 mm)
Window corner radius	1.0 mm
Roof limit	60 deg overhang, 4 mm flat
Open area	about 64% of the band

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 54 (54 windows)
Bore vs reference	identical (windows open wall)
Envelope over reference	+0.003 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	3.8 mm

08

TOPO-FLOW

Continuous undulating contours mimicking wood grain and terrain.

47.8 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	38.3 × 32.1 mm
Peak texture height	3.13 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	38.52 cm³ / 319 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		47.8 g
PETG	1.27 g/cm³		48.9 g
TPU	1.21 g/cm³		46.6 g
ABS	1.04 g/cm³		40.1 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 16.0 m (100 % flow)

DESIGN INTENT

The surface is a contour map of terrain with four wood-like knots. Ridges follow the contours so the lines flow around the handle like grain in a cut log, giving directional grip that is softer on the hand than sharp knurling.

Ridge flanks are rounded and below 45 degrees; the grain is geometry, not colour alone.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- Four knots, contour ridges spaced about 2.5 – 7 mm
- Rounded ridges: crests about 3.1 mm, valleys 1.66 mm above the bore
- Three-tone wood colourway (light / mid / dark)

FILES

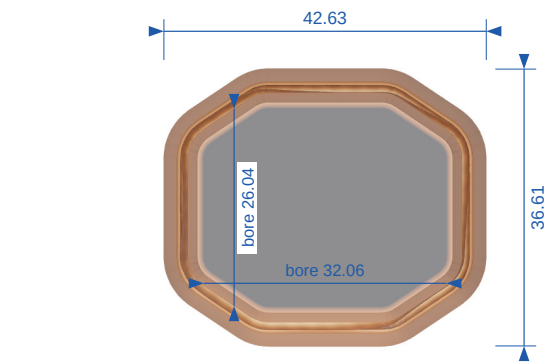
stl/08_topo_flow.stl	19.9 MB · 398,860 triangles
3mf/08_topo_flow.3mf	4.8 MB · colour

✓ FITMENT VERIFIED

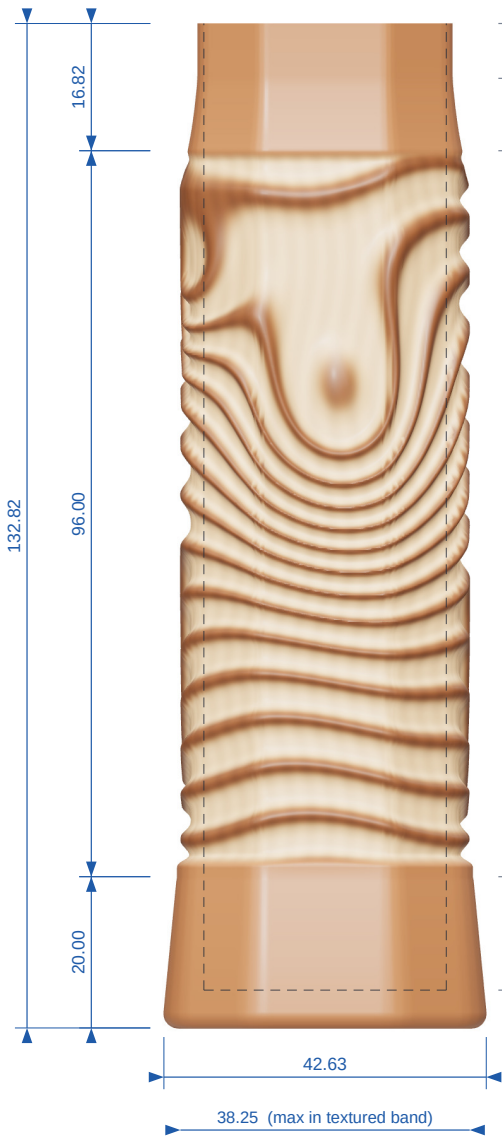
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

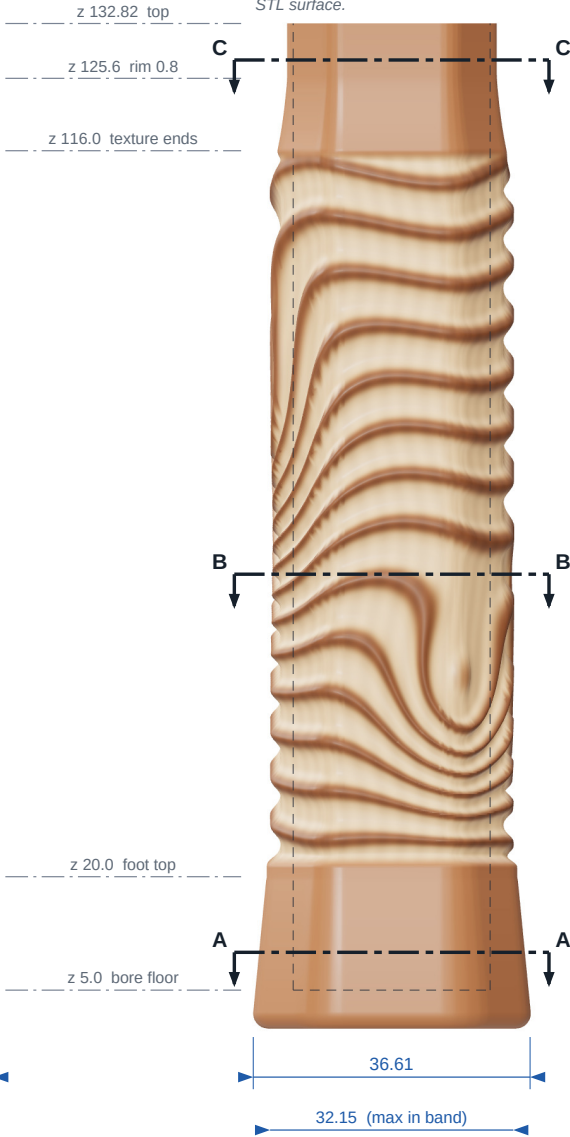
✓ 3MF VALIDATED



TOP VIEW (from +Z)



FRONT VIEW (from -Y)



RIGHT VIEW (from +X)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	38.25 × 32.15
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.13
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.

SCALE 1 : 1



PART

TOPO-FLOW

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

38.52 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

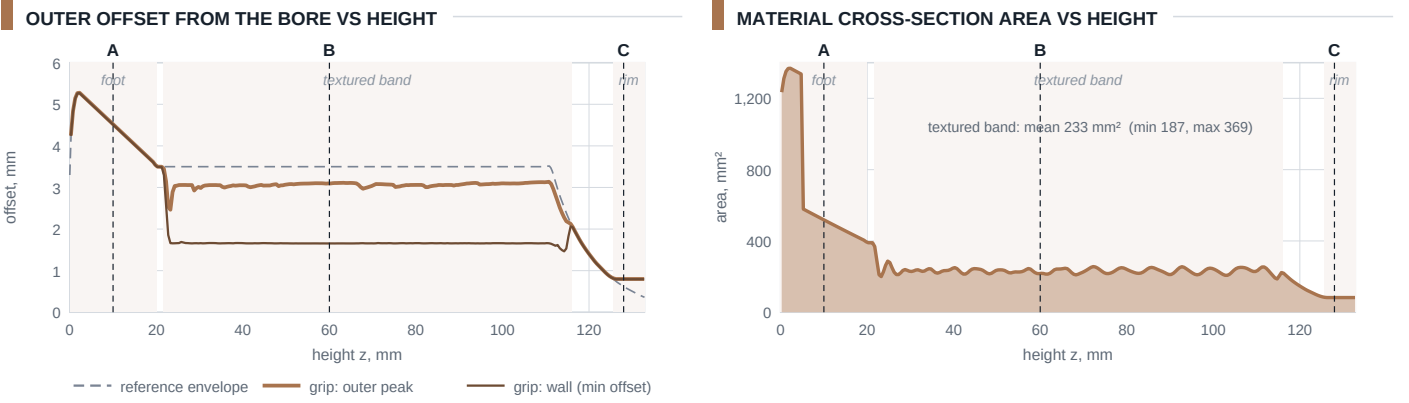
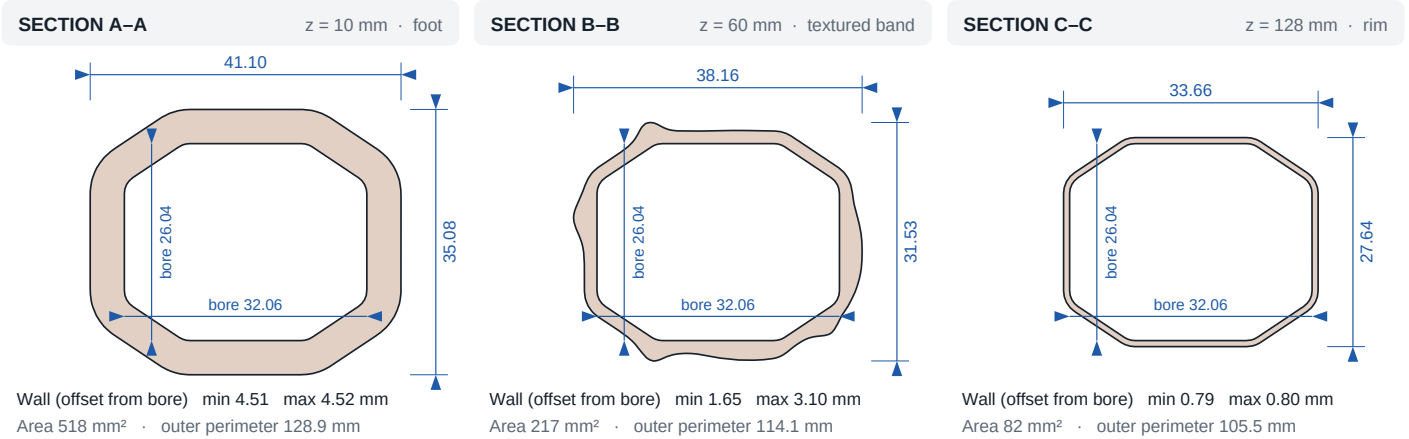
319 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	38.25 × 32.15 mm
Circumference in band	112.2 – 117.8 mm
Volume / surface area	38.52 cm³ / 319.5 cm²

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.65 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			3.13 mm

OUTER DIMENSIONS BY HEIGHT

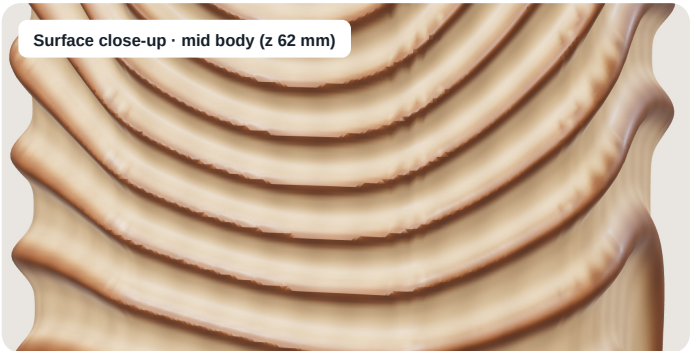
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	37.98	31.02	113.6	229	1.66	3.00
45.0	38.00	32.05	117.4	244	1.66	3.05
60.0	38.13	31.60	114.1	217	1.65	3.10
75.0	37.02	31.88	115.4	220	1.66	3.04
90.0	37.70	31.13	115.4	212	1.66	3.07
105.0	38.18	29.98	116.0	240	1.66	3.11
116.0	36.19	30.19	113.4	221	2.05	2.09
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

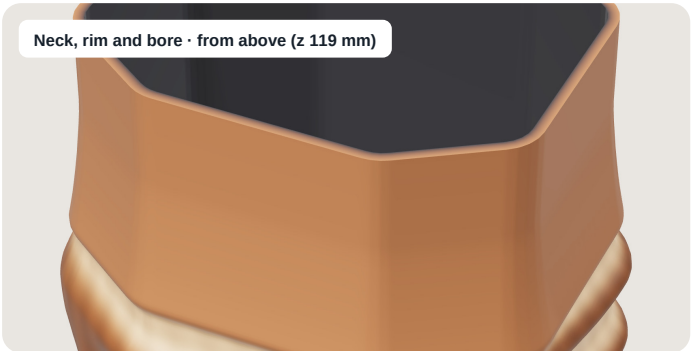
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only

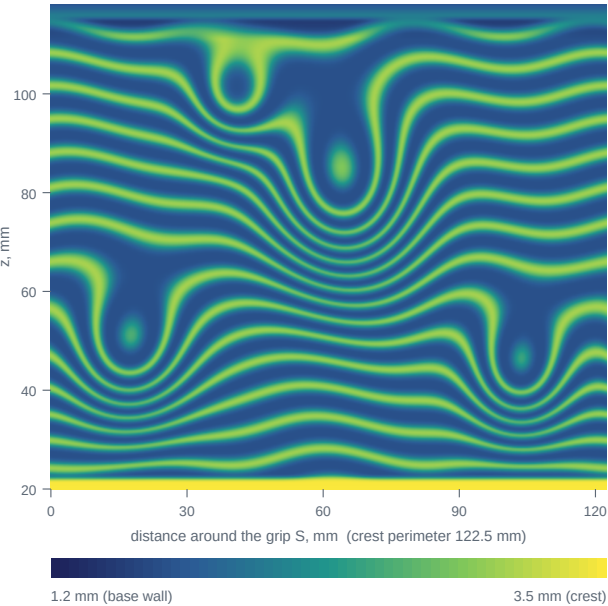


Surface close-up · mid body (z 62 mm)



Neck, rim and bore · from above (z 119 mm)

UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Knots	4
Ridge spacing	about 2.5 – 7 mm
Ridge crest	about 3.1 mm above bore
Valley floor	1.66 mm above bore
Relief	about 1.4 mm
Textured band	z 21.5 – 116.0 mm
Colourway	3 wood tones

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

09

ERGO-CONTOUR

Palm swell, four finger flutes and a thumb rest for hand conformance.

47.0 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	36.9 × 30.9 mm
Peak texture height	2.77 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.89 cm³ / 303 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		47.0 g
PETG	1.27 g/cm³		48.1 g
TPU	1.21 g/cm³		45.8 g
ABS	1.04 g/cm³		39.4 g

50 g limit

COLOURWAY



#424248

3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.8 m (100 % flow)

DESIGN INTENT

A smooth grip shaped for the hand rather than textured: an hourglass swell, a raised palm pad on the back, four finger flutes and a thumb dish on the front. There are no sharp edges anywhere on the surface.

The smoothest surface in the set; it prints with the least visible layer texture.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- Four finger flutes at z = 44, 58, 72, 86 mm
- Thumb dish at z about 97 mm, palm pad at z about 62 mm
- Hourglass swell 1.73 – 2.33 mm, peaking at about 2.8 mm on the palm pad

FILES

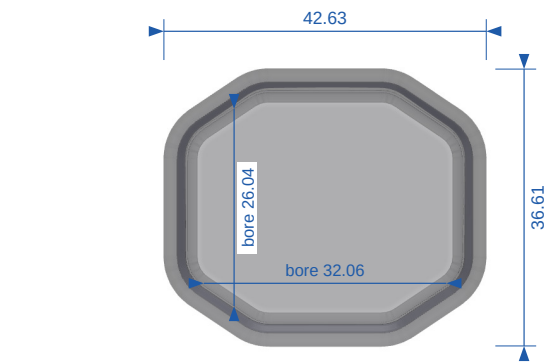
stl/09_ergo_contour.stl	3.4 MB · 68,820 triangles
3mf/09_ergo_contour.3mf	0.8 MB · colour

✓ FITMENT VERIFIED

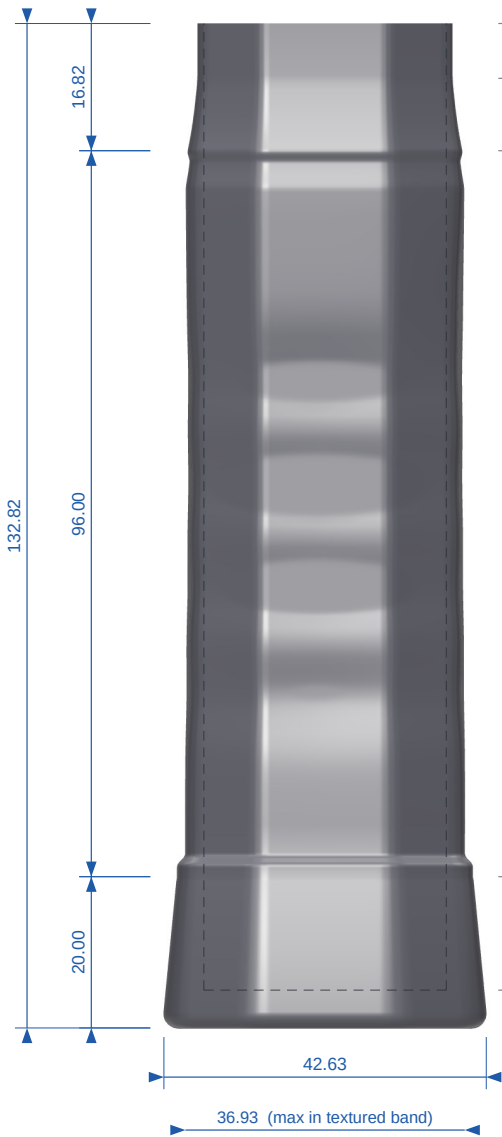
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

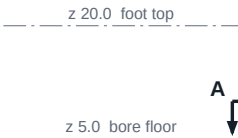
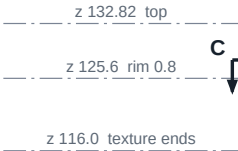
✓ 3MF VALIDATED



TOP VIEW (from +Z)



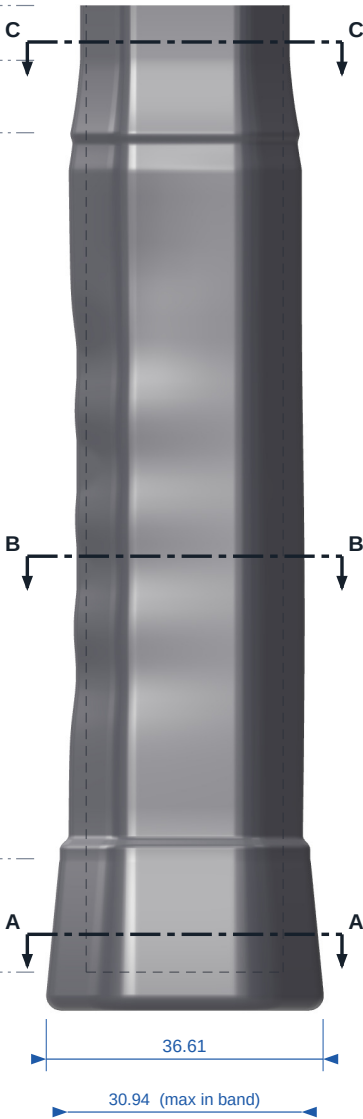
FRONT VIEW (from -Y)



KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	36.93 × 30.94
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	2.77
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

A, B, C: see sheet 3

SCALE 1 : 1



PART

ERGO-CONTOUR

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

MASS (PLA)

47.0 g

REVISION

2

UNITS

millimetres

INFILL

100 %

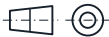
VOLUME

37.89 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

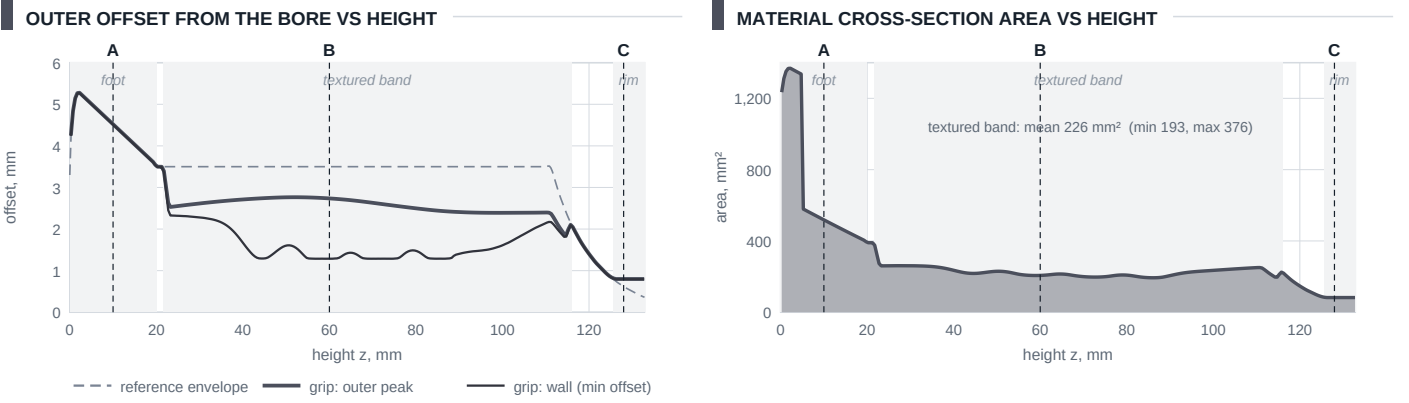
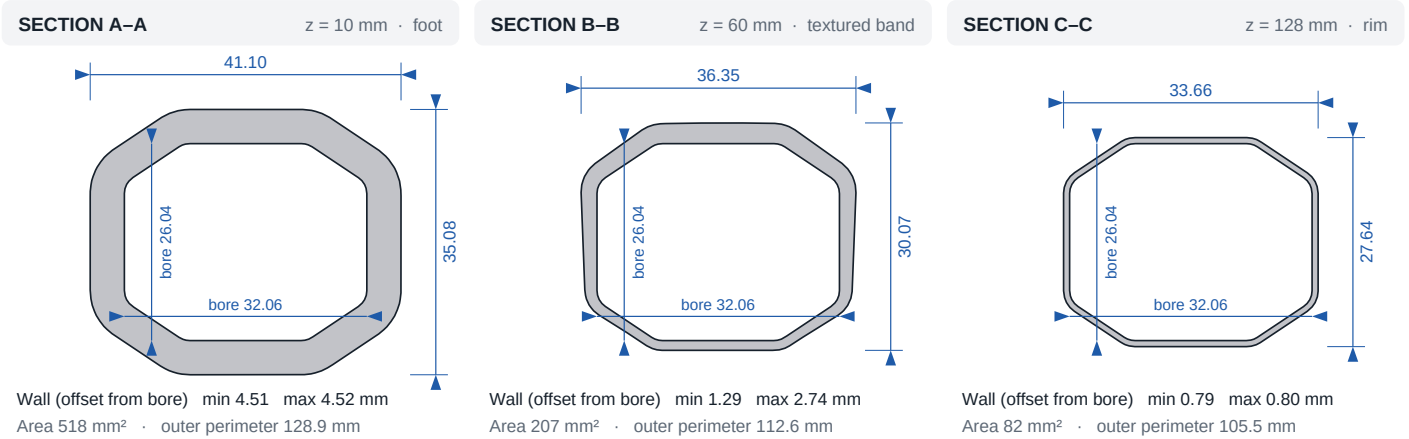
303 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



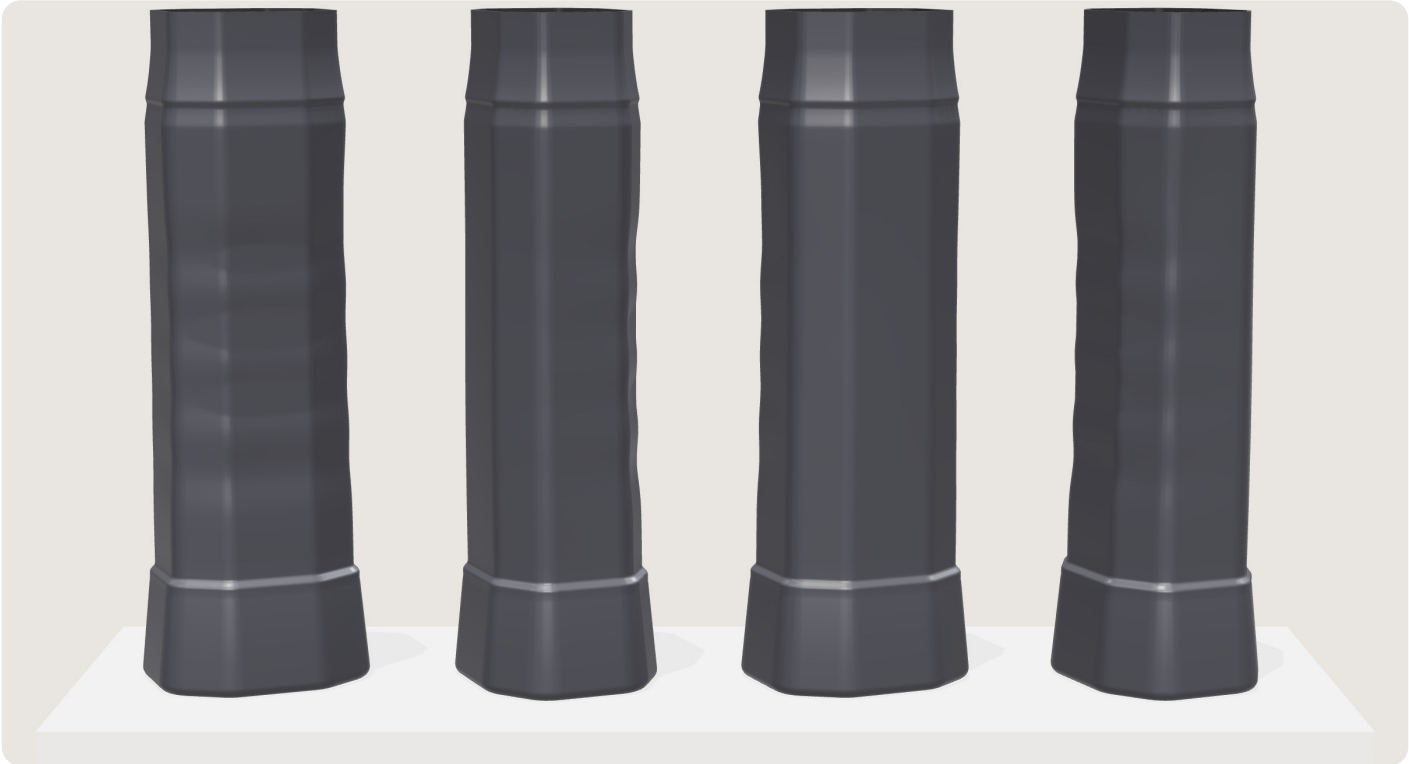
OVERALL & BORE	
Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	36.93 × 30.94 mm
Circumference in band	111.9 – 115.6 mm
Volume / surface area	37.89 cm³ / 303.2 cm²

OUTER DIMENSIONS BY HEIGHT

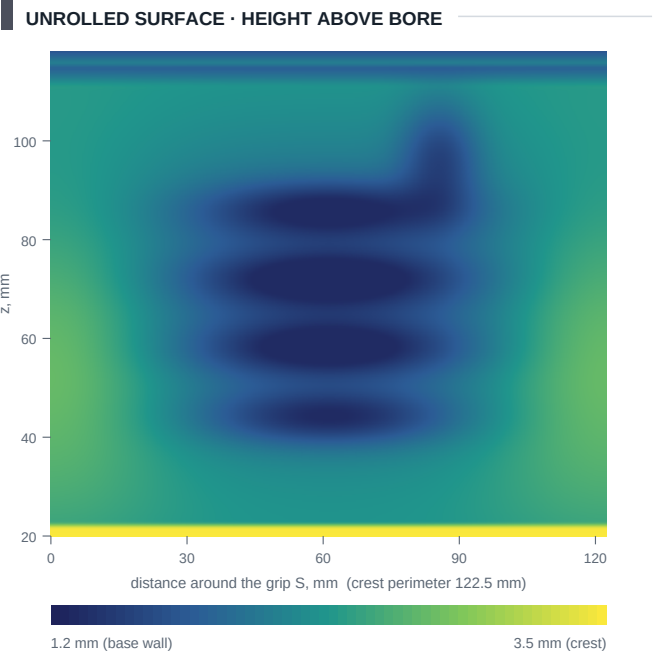
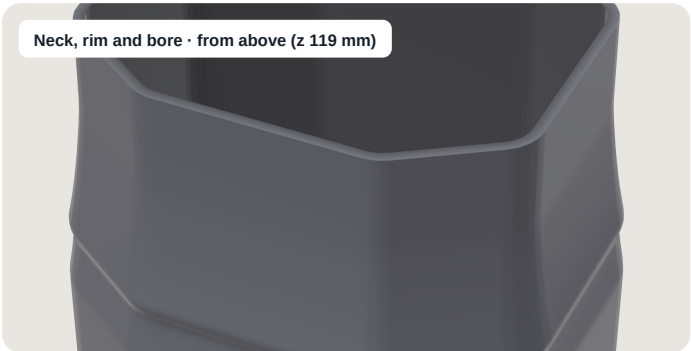
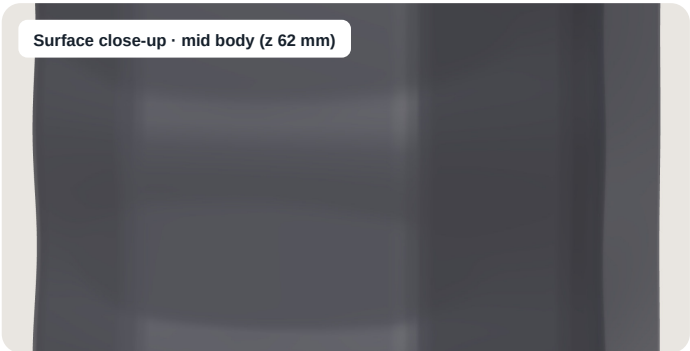
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	36.93	30.94	115.6	261	2.28	2.62
45.0	36.60	30.16	113.3	218	1.30	2.74
60.0	36.35	30.07	112.6	207	1.29	2.74
75.0	36.18	29.90	112.2	199	1.29	2.56
90.0	36.27	30.01	112.5	206	1.40	2.41
105.0	36.66	30.68	114.6	243	1.88	2.39
116.0	36.21	30.19	113.5	222	2.07	2.08
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



DESIGN PARAMETERS	
Finger flutes	4, at z 44 / 58 / 72 / 86
Flute width / depth	about 9 mm / 0.8 mm
Thumb dish	z 97 mm, depth 0.6 mm
Palm pad	back face, z 62 mm, +1.0 mm
Hourglass swell	1.73 – 2.33 mm above bore
Peak height	about 2.8 mm above bore
Surface	smooth, no sharp edges
PRINT & VERIFICATION	
Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

10

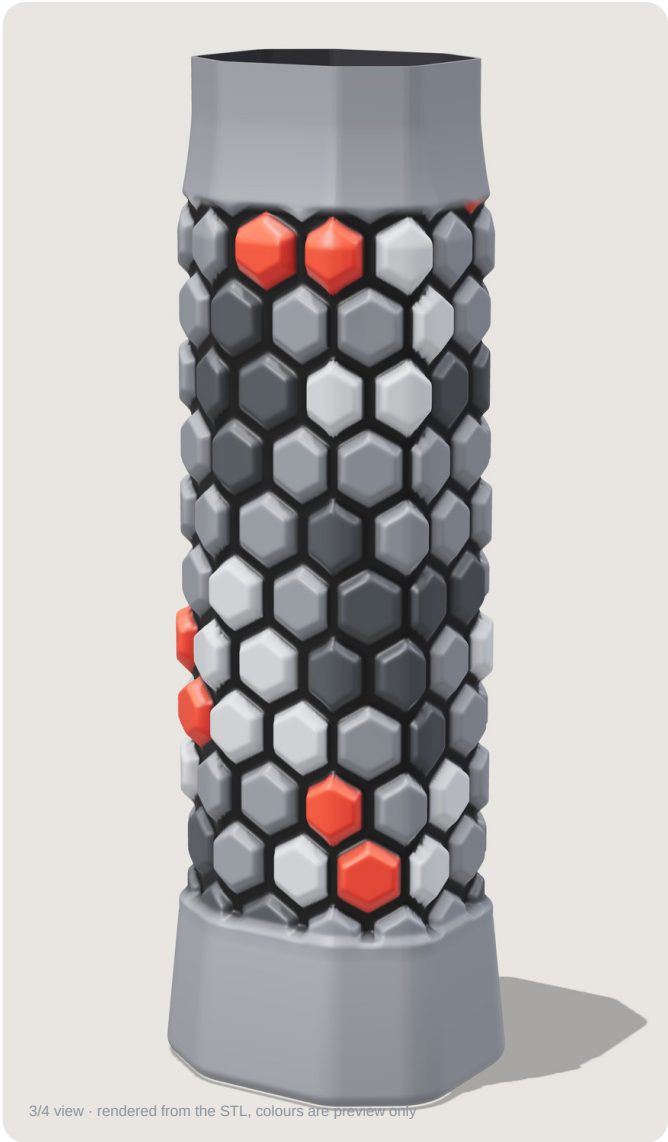
HEXA-MOD

Modular hex pads on four height levels, colour-coded (red = tallest).

47.0 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	38.7 × 33.0 mm
Peak texture height	3.50 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.92 cm³ / 325 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		47.0 g
PETG	1.27 g/cm³		48.2 g
TPU	1.21 g/cm³		45.9 g
ABS	1.04 g/cm³		39.4 g

50 g limit

COLOURWAY

#1C1C1E	#464A4E	#74787E	#B0B4BA	#D63A28

3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.8 m (100 % flow)

DESIGN INTENT

A honeycomb of individual hex pads, each separated by a 1.7 mm gap and set at one of four heights. The height map is pseudo-random with a gentle swell along the handle; the tallest pads (red) reach the full 3.5 mm crest.

Pads are chamfered on the lower edge so they never overhang beyond 45 degrees.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 12 columns × 11 rows of pads, 10.2 mm across flats
- Four levels: 2.35 / 2.76 / 3.13 / 3.50 mm above the bore
- Red pads mark the tallest level (about 6% of the band)

FILES

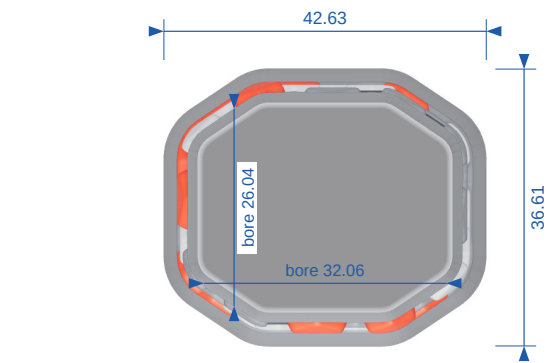
stl/10_hexa_mod.stl	16.2 MB · 324,380 triangles
3mf/10_hexa_mod.3mf	3.8 MB · colour

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

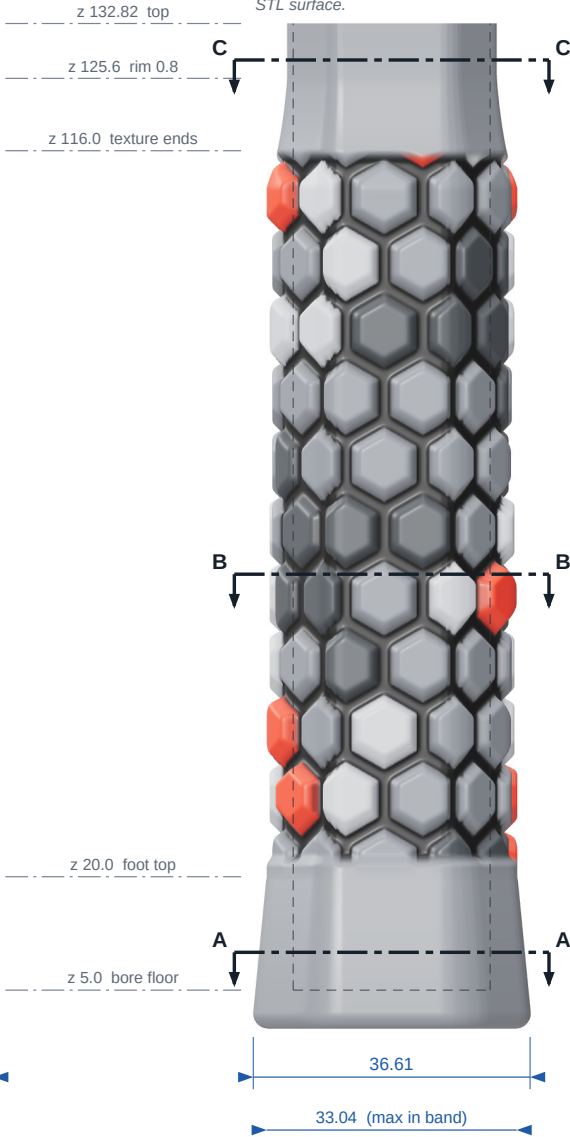
✓ 3MF VALIDATED



TOP VIEW (from +Z)



FRONT VIEW (from -Y)



RIGHT VIEW (from +X)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	38.70 × 33.04
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.50
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.

SCALE 1 : 1



PART

HEXA-MOD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

UNITS

millimetres

PROJECTION



MATERIAL

PLA / PETG / TPU

INFILL

100 %

LAYER HEIGHT

0.20 mm

MASS (PLA)

47.0 g

VOLUME

37.92 cm³

SURFACE AREA

325 cm²

REVISION

2

DATE

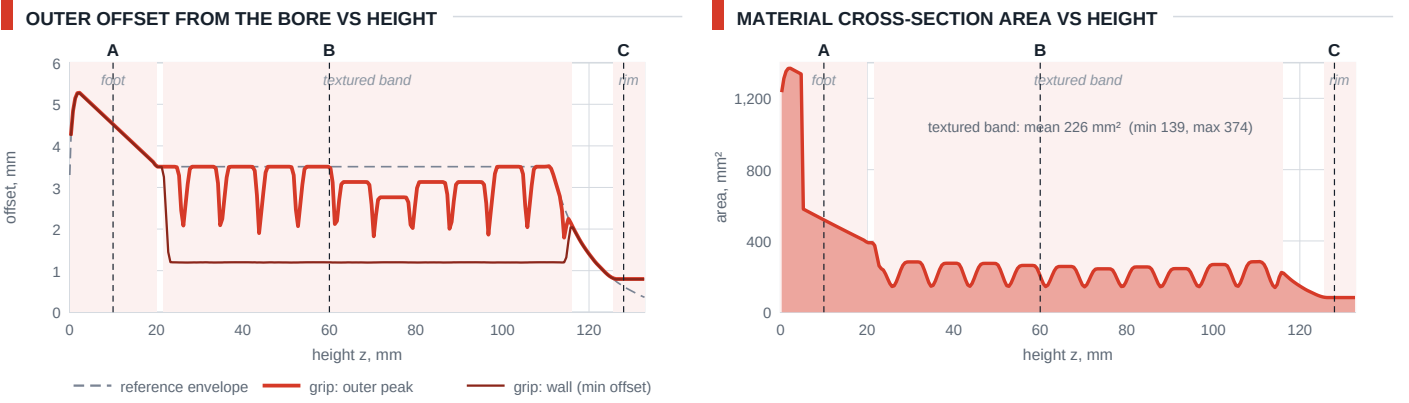
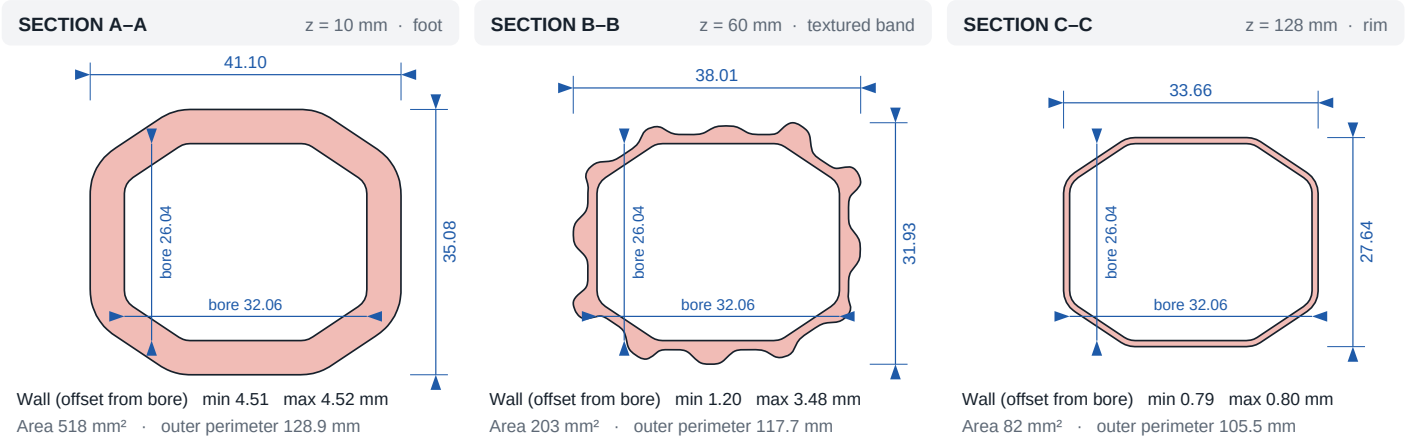
2026-09-30

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	38.70 × 33.04 mm
Circumference in band	110.3 – 119.6 mm
Volume / surface area	37.92 cm ³ / 325.3 cm ²

ZONES BY HEIGHT

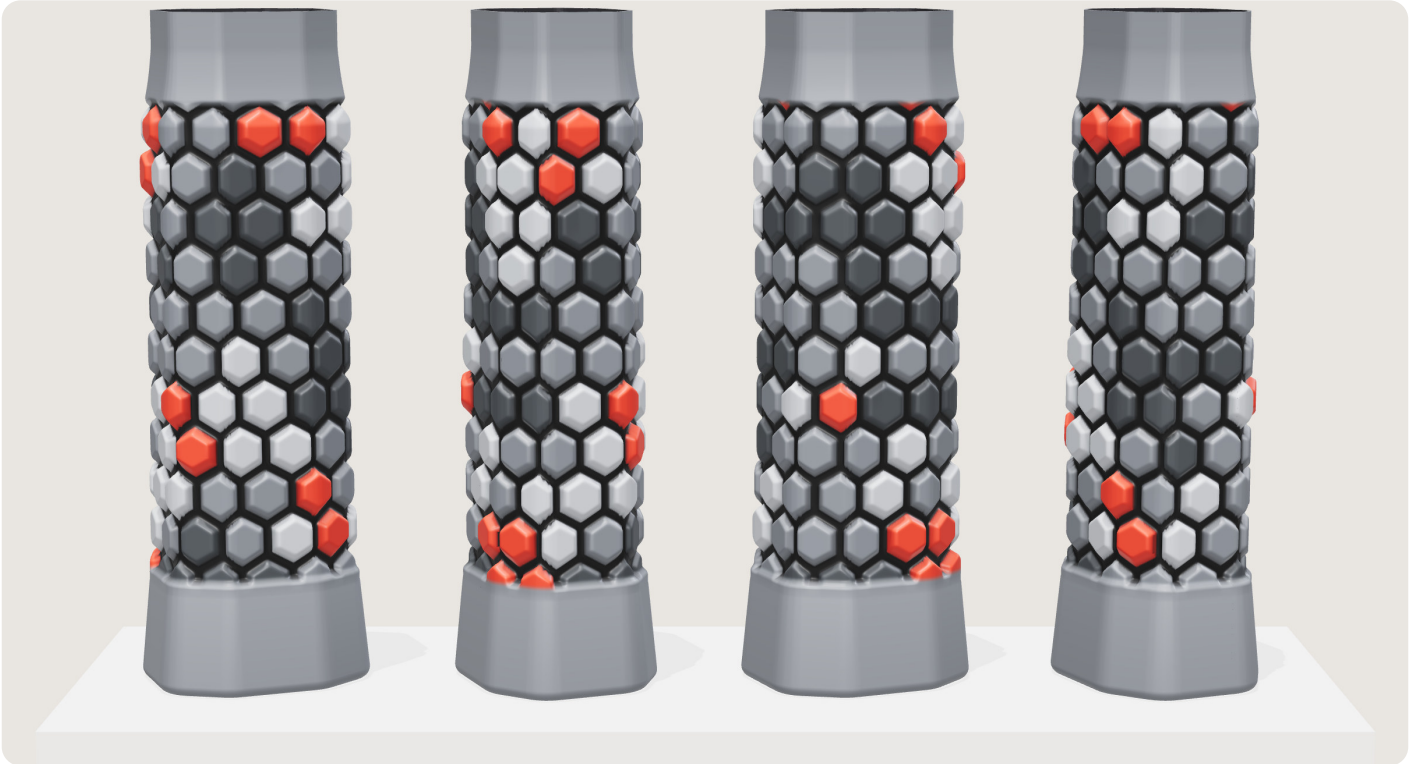
ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.19 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			3.50 mm

OUTER DIMENSIONS BY HEIGHT

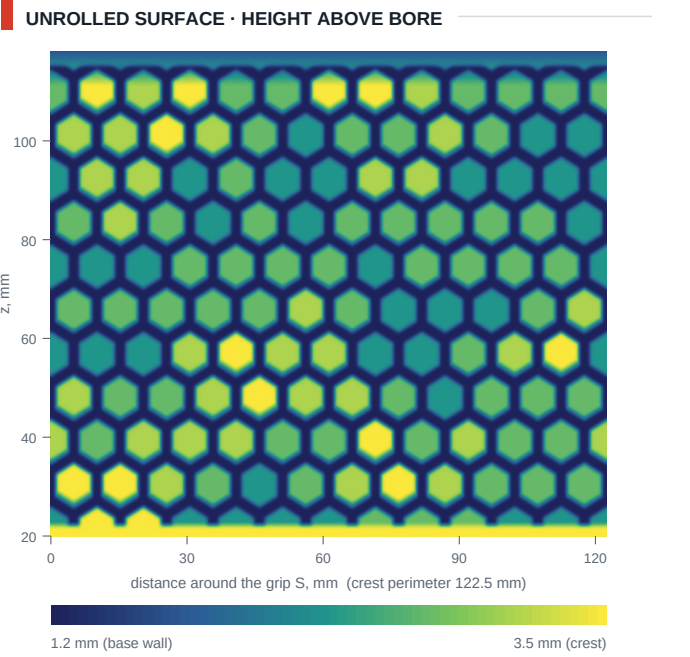
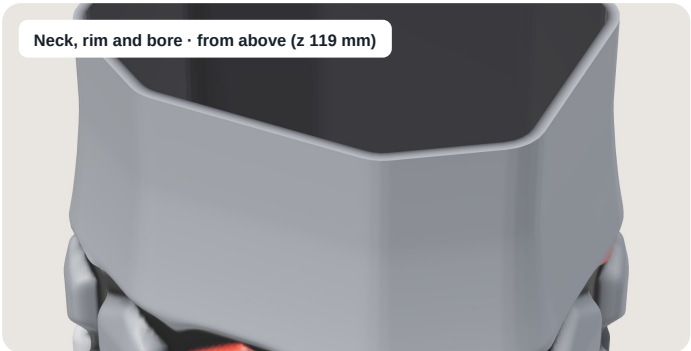
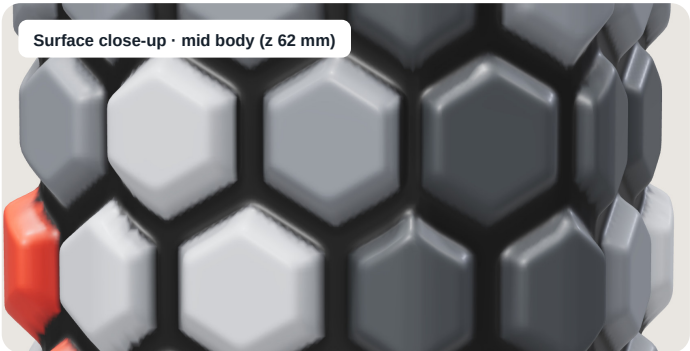
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	38.33	32.67	118.8	282	1.20	3.50
45.0	37.80	32.03	115.1	207	1.20	3.13
60.0	38.07	31.91	117.6	203	1.20	3.48
75.0	37.59	31.49	117.0	244	1.20	2.76
90.0	37.15	32.20	117.7	229	1.20	3.13
105.0	37.70	31.09	112.9	163	1.20	2.94
116.0	36.26	30.24	113.4	221	2.04	2.10
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



DESIGN PARAMETERS	
Pad grid	12 × 11 (pointy-top hex)
Pad across flats	10.21 mm
Gap between pads	1.7 mm
Levels	2.35 / 2.76 / 3.13 / 3.50 mm
Gap floor	1.20 mm (base wall)
Pad edge	chamfered, about 0.8 mm
Level colours	3 greys + red
PRINT & VERIFICATION	
Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none

Notes, method and file manifest

What the numbers mean, what was assumed, and what is in the package.

ABOUT THESE SHEETS

- All dimensions and volumes are measured directly from the STL files; design-intent values are only used for descriptions.
- Weight = volume × density at 100 % infill. Slicers land within a few percent, so budget ±3 %.
- Assumed: 0.4 mm nozzle, 1.75 mm filament, printed upright with the butt on the bed. Not sliced or printed yet.
- Renders and colours are a preview. The STLs are single-material; the 3MFs carry a small print palette per triangle.
- Sheet numbering: PG-01 to PG-10. Drawings are third-angle, true size at 100 % print scale.

REVISION HISTORY

REV	CHANGE
1	Initial ten designs, STL + colour 3MF
2	Smooth finish, solid 1.2 mm walls, ≤ 50 g each, specification sheets

DENSITIES USED (G/CM³)

PLA	1.24
PETG	1.27
TPU	1.21
ABS	1.04

FILES

stl/	ten single-material STLs
3mf/	ten colour 3MFs (materials extension)
specs/	ten sheets + this book
renders/	catalog and fitment overlay
scripts/	generator, verifiers, PDF builder

FILE MANIFEST

NO.	DESIGN	STL	SIZE	TRIANGLES	3MF	SIZE
PG-01	Vortex Grip	01_vortex_grip.stl	19.9 MB	398,860	01_vortex_grip.3mf	4.5 MB
PG-02	Cellular Mod	02_cellular_mod.stl	14.7 MB	294,052	02_cellular_mod.3mf	3.3 MB
PG-03	Tessel-Block	03_tessel_block.stl	19.9 MB	398,860	03_tessel_block.3mf	4.6 MB
PG-04	Logic-Grip	04_logic_grip.stl	19.8 MB	396,900	04_logic_grip.3mf	4.6 MB
PG-05	Neuro-Tread	05_neuro_tread.stl	19.9 MB	398,860	05_neuro_tread.3mf	4.8 MB
PG-06	Carbon Matrix	06_carbon_matrix.stl	19.9 MB	398,860	06_carbon_matrix.3mf	4.7 MB
PG-07	Voronoi-Core	07_voronoi_core.stl	9.7 MB	194,162	07_voronoi_core.3mf	2.3 MB
PG-08	Topo-Flow	08_topo_flow.stl	19.9 MB	398,860	08_topo_flow.3mf	4.8 MB
PG-09	Ergo-Contour	09_ergo_contour.stl	3.4 MB	68,820	09_ergo_contour.3mf	0.8 MB
PG-10	Hexa-Mod	10_hexa_mod.stl	16.2 MB	324,380	10_hexa_mod.3mf	3.8 MB

LIMITS OF THIS DOCUMENT

These sheets describe the geometry and its measured properties. They are not a print guarantee: material, printer, cooling and slicer settings all affect the result. Do a preview slice before committing filament. The reference sleeve remains the authority for fit on your paddle.